

Form 60E

(April 2005)

FORMCODE: ACTE60

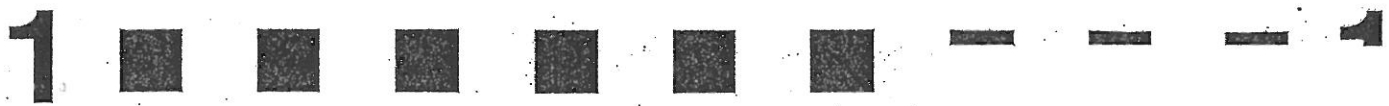
ACT Assessment

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In response to your recent request for test information release materials, this booklet contains the test questions and conversion tables used in determining your ACT scores. Enclosed with this booklet is a report listing your answers to the ACT Assessment tests and the answer key.

If you wish to order a photocopy of your answer document—including, if you took the Writing Test, a copy of your written essay—please use the order form on the inside back cover of this booklet.

We hope that you will find this information helpful.



ENGLISH TEST

45 Minutes—75 Questions

DIRECTIONS: In the five passages that follow, certain words and phrases are underlined and numbered. In the right-hand column, you will find alternatives for the underlined part. In most cases, you are to choose the one that best expresses the idea, makes the statement appropriate for standard written English, or is worded most consistently with the style and tone of the passage as a whole. If you think the original version is best, choose "NO CHANGE." In some cases, you will find in the right-hand column a question about the underlined part. You are to choose the best answer to the question.

You will also find questions about a section of the passage, or about the passage as a whole. These questions do not refer to an underlined portion of the passage, but rather are identified by a number or numbers in a box.

For each question, choose the alternative you consider best and fill in the corresponding oval on your answer document. Read each passage through once before you begin to answer the questions that accompany it. For many of the questions, you must read several sentences beyond the question to determine the answer. Be sure that you have read far enough ahead each time you choose an alternative.

PASSAGE I

Parachuting

You do what you are told. You have had plenty of training. You strap on the equipment and squeeze on the helmet. You follow your instructor and the other two first-time jumpers into the big twin-propeller plane and take your assigned place in front of them all. You will be the first one to go.

The plane has been stripped of all its seats, so everyone is kneeling on the bare metal floor on their knees. The gigantic

engines, which on either side catch and roar

into action. You now can entertain any

second thoughts, the plane has taxied and taken

1. A. NO CHANGE
B. down on their knees.
C. resting on their knees.
D. OMIT the underlined portion and end the sentence with a period.
2. F. NO CHANGE
G. engines, where
H. engines
J. engines that
3. A. NO CHANGE
B. You
C. Then you
D. Before you.
4. F. NO CHANGE
G. took
H. had taken
J. had took

off and is circling upward into the huge cobalt sky. Its
5

headed to an altitude of three thousand feet.
6

It is impossible to talk—or protest—over the
engines' roar; and the noise of the wind that blasts through
7
the open doorways on either side of the narrow fuselage.

8 But your instructor smiles confidently. She gives you
the thumbs-up sign.

[1] Despite your instincts, you fly chest first out
the door of the plane. [2] When the proper altitude is
reached and the signal is given, you crawl to the door.
[3] As taught, you climb into the open doorway and stand
so that your body is half in and half out of the plane, the
wind slamming against you, the sweat soaking the rented
jumpsuit you're wearing. [4] The instructor signals.
[5] Despite your instincts, you nod. 9

You panic. You scream, though no one can
hear you. You think, you might become ill.
10

Furthermore, only seconds pass before the static line,
11
which is a rope attached to you and to the floor of the
plane, becomes tight and opens your chute automatically.
12
You feel like you are soaring upward, though actually

5. A. NO CHANGE
B. Its'
C. It is
D. It was
6. F. NO CHANGE
G. three thousand feet up in altitude.
H. an altitude of three thousand feet up.
J. an altitude of three thousand feet up in the air.
7. A. NO CHANGE
B. roar and the
C. roar. As the
D. roar. The
8. The writer wants to add a sentence to emphasize the narrator's growing sense of nervousness at this point in the essay. Which of the following sentences would best achieve this effect?
F. The wind rushing in is loud.
G. This is just as well, because if you could protest, you might.
H. You keep your mouth shut.
J. You wonder whether the other first-time jumpers are having as good a time as you are.
9. For the sake of unity and coherence, Sentence 1 should be placed:
A. where it is now.
B. after Sentence 3.
C. after Sentence 4.
D. after Sentence 5.
10. F. NO CHANGE
G. think, that you
H. think you
J. think, you then
11. A. NO CHANGE
B. However,
C. Thus
D. Consequently,
12. The writer wants to emphasize the violent action of the parachute opening at this point in the essay. Which of the choices would most successfully achieve this effect?
F. NO CHANGE
G. tightens and opens
H. becomes taut and yanks open
J. quickly becomes taut and opens

1 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 1

you have only slowed down.

¹³
You are sitting in a snug little harness. All is quiet
as you begin to drift gently in the great blue sky and to
appreciate your accomplishment. Thus you are dangling

¹⁴
half a mile above the earth! 15

13. A. NO CHANGE
B. only, slowed down.
C. only slowed down?
D. only slowed, down!

14. F. NO CHANGE
G. When you
H. Since you
J. You

15. The writer wants to end the essay with a sentence that indicates that the narrator feels relaxed floating to earth. Which of the following sentences would most successfully achieve this effect?
- A. You realize with relief that jumping from a plane was a fine idea after all.
B. You are relieved that the jump went well, but you still worry about landing.
C. Well, you jumped.
D. You are fine until the next time you have to jump from a plane.

PASSAGE II

Harmonica Blues

For as long as I can remember, it is true that
¹⁶
I have wanted to play the harmonica. My father
played a few tunes marginally well. I often
asked him to teach me to play, but I quickly grew
impatient, with the wheezing, tinny notes that I

¹⁷
produced. After two weeks' effort, which I would

¹⁸
always set the harmonica aside and had went back
¹⁹
to the baseball field.

My father was able to teach me a few
things about the harmonica. For instance, he
told me that it originated in southern China around
1100 B.C., this early version, the sheng, consisted of long
²⁰
reeds placed in water in a wooden bowl. At one time the

16. F. NO CHANGE
G. it is a fact that
H. it has always been the case that
J. OMIT the underlined portion.

17. A. NO CHANGE
B. impatient with the wheezing, tinny, notes
C. impatient with the wheezing, tinny notes
D. impatient, with the wheezing tinny notes,

18. F. NO CHANGE
G. then
H. in which
J. OMIT the underlined portion.

19. A. NO CHANGE
B. would of gone
C. go
D. would of went

20. F. NO CHANGE
G. B.C. This early version, the sheng, consisted
H. B.C. An early version, the sheng, consisting
J. B.C., the early version, the sheng, consisted

1 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 1

harmonica, which has its origins in ancient China,
²¹
was often confused with the *armonica*, a musical
instrument invented by Benjamin Franklin that
consisted of glasses played by rubbing the rims
with a wet finger. [22] While the harmonica

was marketed originally as a child's
²³

toy, it eventually became a valued portable
²⁴
instrument for folk and blues musicians.

[25] Then, at age fifteen, I decided to enroll
in a blues harmonica class. On the first night, the

instructor, whom had performed in local rock bands
²⁶
for years, guided the class through a tortured version
of "Mary Had a Little Lamb."

21. A. NO CHANGE
B. instrument I have been unable to play, which is called the harmonica,
C. harmonica
D. harmonica, which my father has repeatedly tried to teach me to play,
22. Given that all of the following sentences are true, which one should be added at this point to best continue the historical sketch of the harmonica?
F. The harmonica as we know it was not introduced until 1828 in London.
G. London, where the harmonica was introduced, is a foggy city frequented by tourists who enjoy visiting Big Ben and Buckingham Palace.
H. The most difficult thing about playing the harmonica is pulling air in through the holes; blowing out is easy.
J. Some of Benjamin Franklin's more famous inventions include the Franklin stove and the lightning rod.
23. Which of the following placements for the underlined portion would be LEAST acceptable?
A. Where it is now
B. After the word *harmonica*
C. After the word *was*
D. After the word *as*
24. F. NO CHANGE
G. toy;
H. toy
J. toy, that
25. Which of the following sentences would best provide a logical transition from the preceding paragraph to the new subject in this paragraph?
A. Despite this accumulation of harmonica lore, I still couldn't play.
B. There you have some of the history of the harmonica.
C. As can be seen, I had always been interested in the harmonica.
D. As the reader can see, the harmonica isn't the simple children's toy we so often think it is.
26. F. NO CHANGE
G. whom once
H. who had
J. who did

[1] Then he played the same

melody, but with such a pure, sad wail that,

27

the entire class sat in stunned silence when he'd

finished. [2] When we had finished, he said, "That was

'Mary Had a Little Lamb.'" [3] He picked up his own

harmonica. [4] "This, however, is 'Mary Had the Blues.'" [5] He lifted the instrument to his lips. 28

I returned home determined to learn to play well.

I worked to find the right mouth holes and if I could bend

29

the right notes. All week I wheezed through versions of

"Jingle Bells" and "Oh, Susanna." Finally, on the night

before our next class, which I believe was a Wednesday,

30

in the middle of an attempt at "Old Folks at Home," I

pursed my lips and blew a sweet, piercing note that filled

the room with lovely sound. I had finally played a proper

note on the harmonica, and maybe I'll someday play

the blues.

PASSAGE III

Shaking Up the Language

It's usually hard to determine who first

uttered out loud a given word and almost as hard

31

to know who first wrote it down originally. It is

32

known, however, that William Shakespeare's plays

immortalized many of the words and expressions

popular in his day. Have you ever said you were

33

27. A. NO CHANGE

B. melody, but with such a pure, sad wail that

C. melody, but with such a pure sad wail that,

D. melody but, with such a pure sad wail, that

28. For the sake of the unity and coherence of this paragraph, Sentence 1 should be placed:

F. where it is now.

G. after Sentence 2.

H. after Sentence 3.

J. after Sentence 5.

29. A. NO CHANGE

B. and

C. to see if I were to

D. and if I can

30. F. NO CHANGE

G. which as I recall was a Wednesday,

H. on a Wednesday,

J. OMIT the underlined portion.

31. A. NO CHANGE

B. spoke

C. said verbally

D. gave vocalization to

32. F. NO CHANGE

G. in the beginning.

H. initially.

J. OMIT the underlined portion and end the sentence with a period.

33. A. NO CHANGE

B. you, ever said you were

C. you ever said, you were

D. you ever said you were,

tongue-tied? Do you suspect foul play, bid *good riddance*,
 or felt like a ³⁴*laughingstock*? If so, you were using words
 and phrases that Shakespeare's plays have made
 enduringly popular.

Shakespeare's *Hamlet*, one of his most widely
 read and performed plays, contains many expressions
 that are frequently used today, though often in new ways.
 Hamlet's regard for humanity is evident in his line "What
 a piece of work is a man!" Nonetheless, we still use the
 expression *piece of work*,³⁵ though now it's used to describe
 someone who is hard to understand or deal with. 36

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Iago, the villain of the tragedy, *Othello* strives
 to undo Othello, a respected general. Iago falsely hints
 that Desdemona, the wife of Othello, is in love with
 one of Othello's³⁷ officers. To further confuse Othello,
 Iago warns him against being consumed by doubts³⁸
 about his wife, calling jealousy a "green-eyed monster,"
 a now-common term for that emotion.

The Globe audience heard hundreds of other
 expressions that are now *household words*. In *Henry V*,
 Shakespeare used this very term to describe how the
 names of King Henry's war heroes became as well known
 to English people as simple words used in any household.³⁹

34. F. NO CHANGE
 G. (Do NOT begin new paragraph) Have you sus-
 pected
 H. (Begin new paragraph) Do you suspect
 J. (Begin new paragraph) Have you suspected.
35. A. NO CHANGE
 B. Otherwise, we
 C. While we
 D. We
36. The writer is considering deleting the second clause in
 the preceding sentence, ending the sentence with a
 period after the word *work*. If the writer did this, the
 paragraph would primarily lose:
 F. an example of how an expression changes in
 meaning over time.
 G. an irrelevant digression that undermines the para-
 graph's logic.
 H. information providing a key transition to the next
 paragraph.
 J. a detail that contradicts the statement in the first
 part of this sentence.
37. A. NO CHANGE
 B. tragedy, *Othello*,
 C. tragedy *Othello*,
 D. tragedy *Othello*
38. F. NO CHANGE
 G. one of Othello's
 H. one, of Othello's
 J. one of Othello's
39. A. NO CHANGE
 B. had become
 C. have become
 D. becoming.

1 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 1

A recent film version of *Henry V* was quite popular. Still

40

other terms that his plays have preserved are including

41

birthplace, downstairs, eyeball, and love letter.

We also find illusions of Shakespeare's words

42

in film. For instance, one of the *Star Trek* movies are

43

subtitled "The Undiscovered Country," a phrase from

one of Hamlet's speeches. Given Shakespeare's

strong presence, in speech and popular culture

44

it seems likely that, even in the future,

he will continue to be a popular author.

45

40. F. NO CHANGE
G. The 1989 film adaptation of *Henry V* is itself well known to English and American audiences alike.
H. Those war heroes are colorfully depicted in a recent film version of *Henry V*.
J. OMIT the underlined portion.

41. A. NO CHANGE
B. included:
C. include
D. OMIT the underlined portion.

42. F. NO CHANGE
G. illusions to
H. allusions of
J. allusions to

43. A. NO CHANGE
B. movies is
C. movies,
D. movies, which is

44. F. NO CHANGE
G. presence in speech, and popular culture
H. presence in speech and popular culture,
J. presence, in speech, and popular culture

45. The writer wishes to conclude the sentence and the essay by referring to a main theme of the essay. Which of the following choices would best accomplish that goal?
A. NO CHANGE
B. readers will still be discussing and debating his fascinating ideas.
C. there will be no escaping his influence on the English language.
D. films will continue to borrow plot ideas from his plays.

PASSAGE IV

Maya Lin: Her Work

To the surprise of many, it was a college

46

student, Maya Lin who won the 1981 competition to

47

design the Vietnam Veterans Memorial. Although

Lin's nonrepresentational design initially stirred up a

controversy, the completed monument has won widespread

46. F. NO CHANGE
G. She really surprised many,
H. It was a surprise,
J. It came as a surprise to many,
47. A. NO CHANGE
B. student, Maya Lin,
C. student Maya Lin
D. student Maya Lin,

1 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 1

acclaim for a while now. Visitors to the monument in Washington, D.C., encounter two long black granite walls that meet at an angle as they cut into the ground. Inscribed in the highly polished walls are the names of the more than 58,000 United States military personnel lost in the war.

While some sculptors suggest the magnitude of an event by presenting larger-than-life figures, Lin relies on more understated methods to suggest matters of great significance. [50] Since the unveiling of the Vietnam

Veterans Memorial, Lin has applied this approach in different parts of the country and to different eras of United States history. [52] In 1988, she designed a memorial to honor individuals killed in the civil rights movement. Located in Montgomery, Alabama,

Maya Lin made this work consist of a low granite disk on which the names of the dead appear along with references

48. The writer would like to characterize the monument in general terms to lead into a more detailed discussion of its features. Which choice best accomplishes the writer's goal?

- F. NO CHANGE
- G. from many people.
- H. for Maya Lin's talent.
- J. for its simplicity and power.

49. A. NO CHANGE

- B. figures:
- C. figures.
- D. figures;

50. The writer is considering adding the following sentence at this point in the essay:

After all, it is important to reflect on your life along the way.

Should the writer make this addition here?

- F. No, because it takes a position that is in complete opposition to the spirit of the essay's subject of public sculpture.
- G. No, because it disrupts the essay's tone and expresses an ordinary concept that adds very little to the essay's focus on Maya Lin and her work.
- H. Yes, because it is a generalization that provides readers with a lively way to identify with people who have seen Maya Lin's work.
- J. Yes, because it provides a surprising insight into the role monuments play in public life.

51. A. NO CHANGE

- B. will apply
- C. applied
- D. applying

52. If the writer deleted the phrase "in different parts of the country and" from the preceding sentence, the essay would primarily lose:

- F. a broad claim that is not supported by information elsewhere in the essay.
- G. a broad claim that is supported by details presented later in the essay.
- H. an observation that is out of place in a sentence that focuses on the Vietnam Veterans Memorial.
- J. an observation that undermines claims made elsewhere in the paragraph.

53. A. NO CHANGE

- B. this work consists.
- C. consisting
- D. her work on this monument consisted

to landmark events of the movement. In addition, my research turned up the following information. 54 Behind

the disk stands a granite wall bearing their inscription:

"... UNTIL JUSTICE ROLLS DOWN LIKE WATERS AND
55
 RIGHTEOUSNESS LIKE A MIGHTY STREAM." Lin selected
 these words from a Dr. Martin Luther King Jr. speech,
56
 and in the monument, water flows down the wall, over

Dr. King's words, and out from the center of the disk. 57

Lin continues to challenge herself in a career that combines an appreciation for sculpture, architecture, and the complexity of human experience. A public library in
58
 Ohio, a chapel in Pennsylvania, a train station in New

York—these had been only some of the places that have
59
 been influenced by Maya Lin. It is possible to come away from her work with the sense that the distinctions we sometimes make between public and personal life are blurred in moments of genuine reflection.

54. If the writer deleted the preceding sentence, the essay would primarily lose a:

- F. sense of the writer's attention to the details of Maya Lin's career.
- G. detail about the writer's research that answers a question posed earlier in the essay.
- H. fact that establishes the credentials of the writer as someone with authority on the subject.
- J. reference to the writer of the essay that seems out of place in the absence of other such references.

55. A. NO CHANGE

- B. which
- C. this
- D. it's

56. Given that all the choices are true, which one adds the most specific information to the essay?

- F. NO CHANGE
- G. a speech given by Dr. King in the sixties, specifically 1963,
- H. the "I Have a Dream" speech by Dr. Martin Luther King Jr. himself,
- J. Dr. Martin Luther King Jr.'s 1963 "I Have a Dream" speech,

57. If the opening sentence of this paragraph ("While some sculptors suggest . . .") were deleted, the essay would primarily lose:

- A. a sharp criticism of the shortcomings of works by Maya Lin's colleagues in the field of sculpture.
- B. the only remark that draws an explicit comparison between Maya Lin and other sculptors.
- C. the only reason Maya Lin gives for choosing to work in a fashion unlike that of many other sculptors.
- D. the writer's opinion of who has influenced Maya Lin in her career of designing memorials.

58. F. NO CHANGE

- G. complexity of the nature and character
- H. characteristically complex and inherent complications
- J. complicated character and complex nature

59. A. NO CHANGE

- B. were
- C. are
- D. having been

Question 60 asks about the preceding passage as a whole.

60. Suppose the writer had intended to write an essay focusing on public reaction to the Vietnam Veterans Memorial. Would this essay fulfill the writer's goal?
- F. Yes, because the reaction of the public to the Vietnam Veterans Memorial is described in great detail.
 - G. Yes, because the author clearly avoids limiting the essay to a view of the monument based on first-hand experience.
 - H. No, because the essay describes the monument as part of a broader concern with its creator, who has produced numerous public structures.
 - J. No, because public reaction to the monument is downplayed in the essay in favor of objective information.

PASSAGE V

Zora Neale Hurston, Folklorist

The Harlem Renaissance author Zora Neale Hurston is well known for her novels, especially *Their Eyes Were Watching God*, however, she was also a devoted chronicler

of African American folklore. In her collection of Southern folktales, *Mules and Men*, Hurston, who was born and raised in Eatonville, Florida, suggested her unique qualification for this study when she said she had "the map of Dixie on my tongue."

While she was studying anthropology at Barnard College in New York that Hurston decided to hook up again with the customs, songs, stories, and games she had learned as a child. By her own account, she took the wrong approach to gathering folklore when, in 1927, she first returned to the South.

- 61. A. NO CHANGE
B. even so,
C. but
D. OMIT the underlined portion.
- 62. F. NO CHANGE
G. with
H. for
J. on
- 63. A. NO CHANGE
B. During the time that she was
C. It was while
D. While
- 64. F. NO CHANGE
G. reconnect
H. unite
J. bind
- 65. A. NO CHANGE
B. that,
C. and,
D. OMIT the underlined portion.

1 ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ 1

When she asked them if they knew any folktales, they glanced at her nice ⁶⁶car and "city" clothes, shook their heads, and told her to look elsewhere. On her next trip, she headed straight to her hometown, where she already knew the people and the people knew her.

In Eatonville, her hometown, Hurston went ⁶⁷

to places where she knew the oral tradition would be ⁶⁸thriving. She eagerly sought out the porch of the general

store. ☐ ⁶⁹ As people gathered at the store her first day

there, a wonderful outpouring of tales begun as members ⁷⁰of the crowd vied for their turn to tell stories.

Hurston also visited Polk County, Florida, where she sponsored contests for the most imaginative tall tale.

The winners received prizes, and Hurston collected a wide vastness of folklore. After the contests, people came ⁷¹

66. F. NO CHANGE
G. some of them
H. people
J. OMIT the underlined portion.

67. A. NO CHANGE
B. Eatonville, where she had grown up,
C. her hometown, Eatonville,
D. Eatonville,

68. F. NO CHANGE
G. has been
H. will be
J. is

69. The writer is considering adding the following true statement to the end of the preceding sentence (changing the period after *store* to a comma):

where she had lingered as a child, absorbing what she would later call a "love of talk and song."

Should the writer make this addition at that point?

- A. Yes, because it helps explain why Hurston chose to visit the porch of the general store as part of her research.
B. Yes, because it is the only indication that the general store had an important role in Hurston's work.
C. No, because it has already been established that Hurston was interested in talk and song, making the proposed addition unnecessary.
D. No, because it is irrelevant given the essay's focus on Hurston's work as an adult rather than on her childhood.

70. F. NO CHANGE
G. would begun
H. had began
J. began

71. A. NO CHANGE
B. tremendous prosperity
C. major chunk
D. wealth

to her on their own to share even more stories.

72

Many of the tales Hurston heard during her

Southern trips revealed interesting information:

73

why the woodpecker has a red head, why the alligator is black, or how the snake got its rattles.

Other tales were exaggeratedly and humorous accounts

74

of harsh bosses and smarter workers, or of trickster

animals such as, Brer Rabbit and his cousins. In part

75

because of Hurston's work, these vibrant stories live on.

72. Which of the following alternatives to the underlined portion would NOT be acceptable?

- F. still
- G. many
- H. much
- J. OMIT the underlined portion.

73. Which choice provides the clearest and most precise preview of the tales listed later in this sentence?

- A. NO CHANGE
- B. explained the origins of animal characteristics:
- C. offered important insight into the past:
- D. concerned long-ago events:

74. F. NO CHANGE

- G. exaggeratedly humorously
- H. exaggerated humorously
- J. exaggerated, humorous

75. A. NO CHANGE

- B. as Brer Rabbit
- C. as Brer Rabbit,
- D. as, Brer Rabbit,

END OF TEST 1

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.



MATHEMATICS TEST
60 Minutes—60 Questions

DIRECTIONS: Solve each problem, choose the correct answer, and then fill in the corresponding oval on your answer document.

Do not linger over problems that take too much time. Solve as many as you can; then return to the others in the time you have left for this test.

You are permitted to use a calculator on this test. You may use your calculator for any problems you choose,

but some of the problems may best be done without using a calculator.

Note: Unless otherwise stated, all of the following should be assumed.

1. Illustrative figures are NOT necessarily drawn to scale.
2. Geometric figures lie in a plane.
3. The word *line* indicates a straight line.
4. The word *average* indicates arithmetic mean.

1. What is the value of the expression $(x - 2y)(x + 2y)$ when $x = 4$ and $y = \frac{1}{2}$?

A. 8
 B. 9
 C. 14
 D. 15
 E. 25

2. Tyrone wants to score 80% or better on today's 40-point biology test. What is the least number of points that Tyrone must score to achieve 80%?

F. 4
 G. 8
 H. 20
 J. 32
 K. 36

3. $(2x - 3)(3x - 4)$ is equivalent to:

A. x^2
 B. $5x^2 - 12x - 7$
 C. $5x^2 + 12$
 D. $6x^2 - 17x + 12$
 E. $6x^2 + 12$

4. The probability that an event happens is $\frac{2}{9}$. What is the probability that the event does NOT happen?

F. 0
 G. $\frac{1}{9}$
 H. $\frac{2}{9}$
 J. $\frac{7}{9}$
 K. $\frac{9}{2}$

DO YOUR FIGURING HERE.



DO YOUR FIGURING HERE.

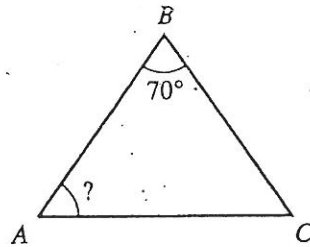
5. A recipe for 150 servings requires 4.5 liters of sauce. About how many liters of sauce are required for 80 servings?

A. 2.1
B. 2.4
C. 3.0
D. 8.4
E. 15.5

6. Pei bought a 36-inch-long candle. It burns at a steady rate of 0.48 inches every hour. Which of the following is a relationship between the length, l , in inches, of the candle and the time, t , in hours, the candle has burned?

F. $l = 36 - 0.48t$
G. $l = 36 + 0.48t$
H. $l = 0.48t$
J. $l = 35.52t$
K. $l = 75t$

7. In $\triangle ABC$ below, $AB = BC$, and $\angle ABC$ measures 70° . What is the measure of $\angle BAC$?



A. 35°
B. 50°
C. 55°
D. 70°
E. 110°

8. You have 4 rectangular regions that you want to cover with cedar chips. One region is 2 feet by 8 feet; another is 1 foot by 4 feet; another is 2 feet by 20 feet; and the last one is 1 foot by 5 feet. If a bag of cedar chips covers 10 square feet, how many bags of cedar chips will you need?

F. 5
G. 6
H. 7
J. 9
K. 12

9. A function that is defined by the set of ordered pairs $\{(2,1), (4,2), (6,3)\}$ has domain $\{2, 4, 6\}$. What is the domain of the function defined by the set of ordered pairs $\{(0,2), (2,2), (3,-2)\}$?

A. $\{2\}$
B. $\{-2, 2\}$
C. $\{-2, 0, 3\}$
D. $\{0, 2, 3\}$
E. $\{-2, 0, 2, 3\}$



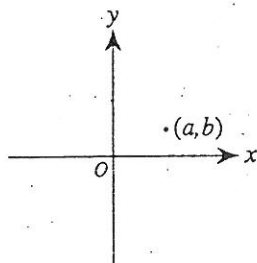
10. Let p and q be statements.

Statement S : If p , then q .
 Converse of S : If q , then p .
 Inverse of S : If not p , then not q .
 Contrapositive of S : If not q , then not p .

One of the following statements is the converse of the statement "If the lights are on, then the store is open." Which one is it?

- F. If the store is open, then the lights are on.
 G. If the lights are not on, then the store is not open.
 H. If the store is not open, then the lights are not on.
 J. The lights are not on.
 K. The store is not open.

11. A point with coordinates (a,b) is plotted in the standard (x,y) coordinate plane as shown below. The point is then reflected across the y -axis. Which of the following are the coordinates for the point after the reflection?



- A. $(-a, b)$
 B. $(a, -b)$
 C. (b, a)
 D. $(-b, a)$
 E. $(b, -a)$

12. What quantity must be added to $4x^2 + x - 5$ to obtain $x^2 + 5x + 1$?

- F. $5x^2 + 6x + 6$
 G. $5x^2 + 6x - 4$
 H. $-3x^2 + 6x + 6$
 J. $-3x^2 + 6x - 4$
 K. $-3x^2 + 4x + 6$

13. A 5-pound bag of potatoes costs \$1.99 at Grocery Garden. A 5-pound bag of the same type of potatoes costs \$2.19 at Food Fair. How much cheaper, *per pound*, are the Grocery Garden potatoes than the Food Fair potatoes?

- A. \$0.02
 B. \$0.04
 C. \$0.20
 D. \$1.00
 E. \$2.00

DO YOUR FIGURING HERE.



14. What is the solution of the equation $12x = -8(10 - x)$?

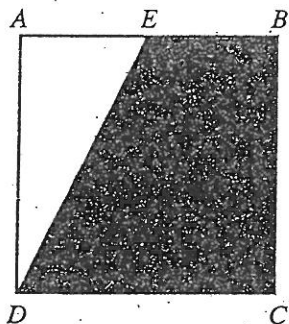
DO YOUR FIGURING HERE.

- F. -20
 G. $-6\frac{2}{13}$
 H. -4
 J. $\frac{2}{13}$
 K. $7\frac{3}{11}$

15. The \$414 price a store charges for a small refrigerator consists of the refrigerator's original cost to the store plus a profit of 15%. What was the refrigerator's original cost to the store?

- A. \$ 54
 B. \$360
 C. \$399
 D. \$429
 E. \$478

16. In the figure below, $ABCD$ is a square with side length 10 centimeters. The midpoint of $\overline{AB}$ is E . What is the area, in square centimeters, of the shaded region?



- F. 25
 G. 50
 H. $50\sqrt{5}$
 J. 75
 K. 100

17. What is the solution to the following system of equations?

$$\begin{aligned} x + 2y &= 12 \\ x - y &= 3 \end{aligned}$$

- | | x | y |
|----|-----|-----|
| A. | -6 | 9 |
| B. | 2 | 5 |
| C. | 3 | 4.5 |
| D. | 6 | 3 |
| E. | 18 | 15 |

18. If $f(x) = 2x$ and $g(x) = x^2$, what is $f(g(3))$?

- F. 6
 G. 12
 H. 18
 J. 36
 K. 54



19. A dinner combination at Nassif's Family Restaurant contains exactly 1 appetizer, 1 main course, and 1 dessert. How many different dinner combinations are possible when choosing from 5 appetizers, 4 main courses, and 5 desserts?

A. 14
B. 40
C. 45
D. 80
E. 100

DO YOUR FIGURING HERE.

20. Which of the following is equivalent to the inequality $-5 + m \leq -4 + 2m$?

F. $m \geq -9$
G. $m \geq -3$
H. $m \geq -1$
J. $m \geq -\frac{1}{3}$
K. $m \geq 3$

21. Which of the following lists orders $\frac{1}{3}$, 0.28, $\frac{2}{5}$, $\frac{3}{8}$, 0.37, and $\frac{3}{10}$ from least to greatest?

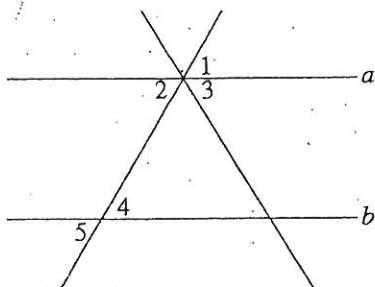
A. 0.28, $\frac{3}{10}$, $\frac{1}{3}$, $\frac{3}{8}$, 0.37, $\frac{2}{5}$
B. 0.28, $\frac{3}{10}$, $\frac{1}{3}$, 0.37, $\frac{3}{8}$, $\frac{2}{5}$
C. 0.28, 0.37, $\frac{3}{10}$, $\frac{1}{3}$, $\frac{3}{8}$, $\frac{2}{5}$
D. 0.28, $\frac{3}{10}$, $\frac{3}{8}$, $\frac{2}{5}$, $\frac{1}{3}$, 0.37
E. $\frac{1}{3}$, $\frac{2}{5}$, $\frac{3}{8}$, $\frac{3}{10}$, 0.28, 0.37

22. Which of the following is equivalent to $(3x^3)^{-2}$?

F. $\frac{1}{9x^6}$
G. $\frac{1}{9x^9}$
H. $\frac{3}{x^6}$
J. $-6x^3$
K. $-9x^6$



23. In the figure below, line a is parallel to line b . The 2 transversals intersect at a point on line a . Which of the following pairs of angles is NOT necessarily congruent?



- A. $\angle 1$ and $\angle 4$
 B. $\angle 1$ and $\angle 5$
 C. $\angle 2$ and $\angle 3$
 D. $\angle 2$ and $\angle 4$
 E. $\angle 4$ and $\angle 5$
24. If $A = 10^{B+C}$, what is $\log_{10} A$?

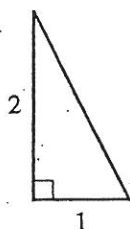
- F. B
 G. $B + C$
 H. 10^A

I. $10^B + 10^C$
 K. 10^{B+C}

25. What is the solution set of the equation $4x^2 - 9 = 0$?

- A. $\{-\sqrt{2}, \sqrt{2}\}$
 B. $\{-\frac{3}{2}, \frac{3}{2}\}$
 C. $\{-\sqrt{3}, \sqrt{3}\}$
 D. $\{-2, 2\}$
 E. $\{\frac{9}{4}\}$

26. The given dimensions of the right triangle shown below are in inches. What is the length, in inches, of the hypotenuse of the triangle?



- F. $\sqrt{3}$
 G. $\sqrt{5}$
 H. 2
 J. 3
 K. 5

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DO YOUR FIGURING HERE.



DO YOUR FIGURING HERE.

27. Pilar obtained estimates for cleaning her furnace from 2 heating companies. Lehman Heating's estimate was \$30 for a service call plus \$22 per hour for cleaning the furnace. A-1 Heating's estimate was \$35 for a service call plus \$20 per hour for cleaning the furnace. If the estimates were the same in both the total amount and the number of hours for cleaning Pilar's furnace, how many hours for cleaning the furnace were reflected in the estimates?

- A. 2
- B. $2\frac{1}{2}$
- C. 3
- D. $3\frac{1}{2}$
- E. 4

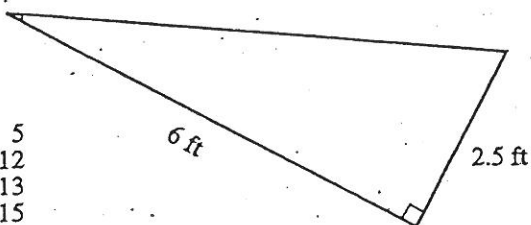
28. In the standard (x,y) coordinate plane, what is the slope of the line that passes through the points (-3,5) and (7,3)?

- F. -5
- G. -2
- H. $-\frac{1}{5}$
- J. $\frac{1}{2}$
- K. 2

29. If 4 is the first term and 256 is the fourth term of a geometric progression, which of the following is the second term?

- A. 8
- B. 12
- C. 16
- D. 32
- E. 64

30. The legs of the right triangle below are doubled in length to become the legs of a new right triangle. What is the length, in feet, of the longest side of the new triangle?



- F. 5
- G. 12
- H. 13
- J. 15
- K. 17



31. If $3x + a = 9$, then, in terms of a , $x = ?$

DO YOUR FIGURING HERE.

- A. $\frac{a}{6}$
B. $3a$
C. $3 - a$
D. $\frac{9-a}{3}$
E. $\frac{9+a}{3}$
32. As part of one day's training workout, 5 members of a track team ran for exactly 3 minutes each. The 5 runners ran the 5 distances given below, in miles. Which distance corresponds to the fastest speed?

- F. $\frac{3}{5}$
G. $\frac{3}{8}$
H. $\frac{5}{8}$
J. $\frac{7}{9}$
K. $\frac{11}{16}$

33. Which of the following is an equation for the line passing through $(0,0)$ and $(4,3)$ in the standard (x,y) coordinate plane?

- A. $x - y = 1$
B. $x + y = 7$
C. $3x - 4y = 0$
D. $3x + 4y = 25$
E. $4x + 3y = 25$

34. Points P , Q , R , and S lie on a line in the order given. Point R is the midpoint of $\overline{QS}$, $\overline{PR}$ is 5 cm long, and $\overline{PS}$ is 7 cm long. How many centimeters long is $\overline{QS}$?

- F. 2
G. 3
H. 4
J. 5
K. 6

35. Which of the following expressions is equivalent to $3x^4 + 6x^2 - 45$?

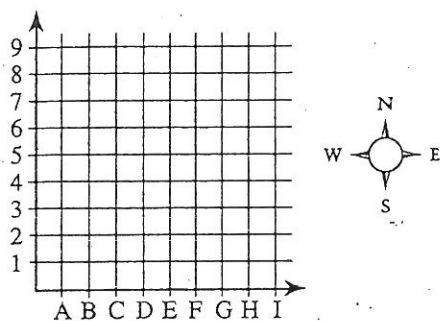
- A. $(x^2 + 5)(x^2 - 3)$
B. $(3x^2 - 15)(x^2 + 3)$
C. $3(x^4 + 6x^2 - 45)$
D. $3(x^2 - 5)(x^2 + 3)$
E. $3(x^2 + 5)(x^2 - 3)$



Use the following information to answer questions 36–38.

DO YOUR FIGURING HERE.

A student is working on a social studies project with her partner. They are drawing local points of interest on a map that is laid out in a coordinate plane, as shown below, where the distance between adjacent grid lines is 1 cm and represents 4 miles.



36. The student tells her partner that 2 high schools are to be $2\frac{1}{2}$ cm apart on the map. About how many miles apart are these high schools?

F. 12
G. 10
H. 9
J. 8
K. 6

37. Two historical landmarks are 24 miles apart. One landmark is at D-3; the second landmark is due northeast of the first one. Which one of the following is closest to the second landmark's location?

A. E-8
B. F-6
C. G-6
D. H-7
E. I-4

38. A square park measures $\frac{1}{2}$ mile on each side. What should be the park's area, in square centimeters, on the map?

F. $\frac{1}{64}$
G. $\frac{1}{16}$
H. $\frac{1}{8}$
J. 1
K. 2



39. A quadrilateral has diagonals that do NOT bisect each other. Which of the following types of quadrilaterals could it be?

A. Parallelogram
 B. Rectangle
 C. Rhombus
 D. Square
 E. Trapezoid

DO YOUR FIGURING HERE.

40. Mary takes 2 medications throughout the day and night. One medication is to be taken every 6 hours and the other is to be taken every 4 hours. If Mary begins taking both medications at 7:00 A.M. and takes both medications on schedule, how many hours later will it be when she next takes both medications at the same time?

F. 6
 G. 9
 H. 10
 J. 12
 K. 24

41. The table below contains information about the 1996 season baseball standings for the American League Eastern Division. Which team won the most games away from home in 1996?

Team	Wins	Losses	Winning rate	Home win-loss record
New York	92	70	.568	49-31
Baltimore	88	74	.543	43-38
Boston	85	77	.525	47-34
Toronto	74	88	.457	35-46
Detroit	53	109	.327	27-54

A. New York
 B. Baltimore
 C. Boston
 D. Toronto
 E. Detroit

42. If 1 is a solution of the equation $x^2 + hx + 10 = 0$, what does h equal?

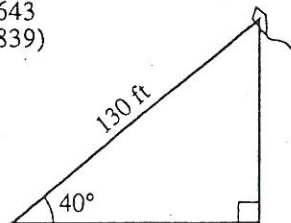
F. -11
 G. -10
 H. -1
 J. 10
 K. 11



43. Benjamin is flying a kite using 130 feet of string, as shown in the figure below. His string makes an angle of 40° with the level ground. About how many feet above the ground is the kite when the string is taut?

(Note: $\cos 40^\circ \approx 0.766$
 $\sin 40^\circ \approx 0.643$
 $\tan 40^\circ \approx 0.839$)

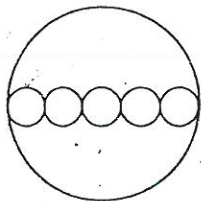
- A. 80
 B. 100
 C. 110
 D. 170
 E. 200



DO YOUR FIGURING HERE.

44. The 5 congruent circles shown in the diagram below have their centers lying on the same diameter of the larger circle. Each circle is tangent to 2 other circles. If the area of each small circle is 4π square units, what is the area, in square units, of the larger circle?

- F. 100π
 G. 50π
 H. 40π
 J. 30.5π
 K. 20π



45. Which of the following is equivalent to $\sqrt[3]{27r^6}$?

- A. $3r^2$
 B. $3r^3$
 C. $9r^2$
 D. $9r^3$
 E. $729r^2$

46. There is a pattern when adding the cubes of the first c consecutive counting numbers, as illustrated below.

$$1^3 + 2^3 = 9 = (1 + 2)^2$$

$$1^3 + 2^3 + 3^3 = 36 = (1 + 2 + 3)^2$$

Which of the following is an expression for the sum of the cubes of the first c consecutive counting numbers?

- F. $(c + 1)^3$
 G. $(c + 1)^2$
 H. $(1 + 2 + \cdots + c)^c$
 J. $(1 + 2 + \cdots + c)^3$
 K. $(1 + 2 + \cdots + c)^2$



47. The volume of a sphere is $\frac{4}{3}\pi r^3$, where r is the radius of the sphere. The shapes of the planets Uranus and Earth are approximately spheres. The radius of Uranus is about 4 times that of Earth. About how many times the volume of Earth is the volume of Uranus?

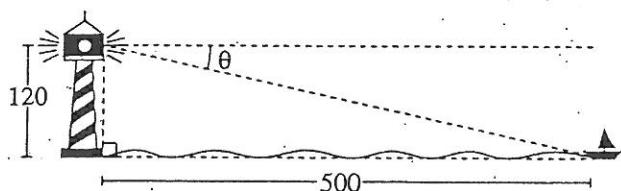
A. 81
B. 64
C. 16
D. 12
E. 4

DO YOUR FIGURING HERE.

48. The 5 positive integers a , a , a , b , and c have an average of a . What is the value of $\frac{b+c}{2}$?

F. $\frac{a}{3}$
G. $\frac{a}{2}$
H. a
J. $2a$
K. $3a$

49. From the top of a 120-foot lighthouse, a small sailboat is sighted on the water 500 feet from the base of the lighthouse, as shown below. The angle of depression, θ , can be found using which of the following equations?



A. $\cos \theta = \frac{120}{500}$
B. $\sin \theta = \frac{500}{120}$
C. $\sin \theta = \frac{120}{500}$
D. $\tan \theta = \frac{500}{120}$
E. $\tan \theta = \frac{120}{500}$



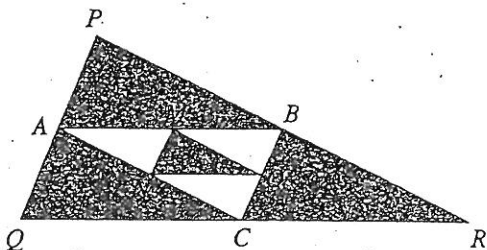
50. A rectangle that measures 4 cm by 6 cm is divided into 24 squares with sides 1 cm in length. What is the total number of 1 cm long sides in those 24 squares?

(Note: If 2 squares share a side, the side should be counted only once.)

- F. 24
G. 48
H. 58
J. 70
K. 96

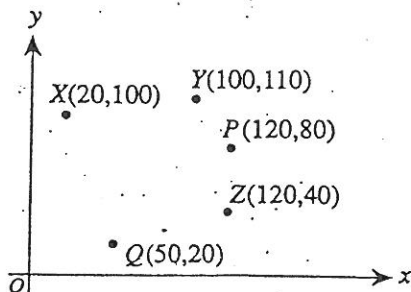
DO YOUR FIGURING HERE.

51. In the figure below, A , B , and C are the midpoints of the sides of $\triangle PQR$, and M , N , and O are the midpoints of the sides of $\triangle ABC$. The interiors of $\triangle PAB$, $\triangle AQC$, $\triangle BCR$, and $\triangle MNO$ are shaded. What fraction of the interior of $\triangle PQR$ is shaded?



- A. $\frac{1}{2}$
B. $\frac{4}{7}$
C. $\frac{5}{8}$
D. $\frac{3}{4}$
E. $\frac{13}{16}$

52. The Malcolm X High School Student Council is planning a spring outdoor carnival. In the figure below, the locations for several planned activities are represented as points X , Y , Z , P , and Q on the standard (x,y) coordinate plane, where each coordinate unit represents 1 yard. The ticket booth is to be placed exactly midway between X and Z . About how many yards will the ticket booth be from Y ?



- F. 30
G. 35
H. 40
J. 42
K. 50



53. For what value of t does the equation $h = 3t - t^2$ have its maximum value for h ?

DO YOUR FIGURING HERE.

- A. $\frac{3}{2}$
- B. 2
- C. $\frac{9}{4}$
- D. 3
- E. $\frac{9}{2}$

54. For all $x > 0$, which of the following is equivalent to

$$\frac{1}{x+1} + \frac{1}{x}?$$

- F. $\frac{1}{x(x+1)}$
- G. $\frac{1}{2x+1}$
- H. $\frac{2}{x(x+1)}$
- J. $\frac{2}{x} + 1$
- K. $\frac{2x+1}{x(x+1)}$

55. The interior angles of a quadrilateral are in the ratio 1:2:3:4. What is the measure of the largest interior angle?

- A. 100°
- B. 108°
- C. 120°
- D. 144°
- E. 160°

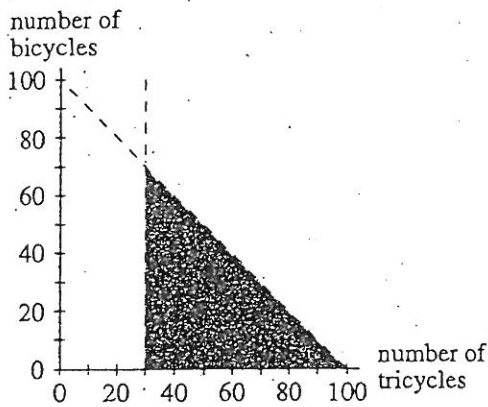
56. The 2 legs of a right triangle measure 37 inches and 45 inches, respectively. What is the cosine of the triangle's smallest interior angle?

- F. $\frac{37}{45}$
- G. $\frac{45}{37}$
- H. $\frac{37}{\sqrt{37+45}}$
- J. $\frac{37}{\sqrt{37^2+45^2}}$
- K. $\frac{45}{\sqrt{37^2+45^2}}$



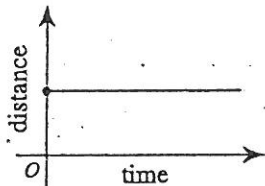
57. In creating a business plan for a cycle shop that sells both tricycles and bicycles, a business analyst graphed the inventory constraints set by the shop owner, and then shaded the region, shown below, where all of the constraints were met. Which of the following statements defines the shaded region?

DO YOUR FIGURING HERE.



- A. The number of tricycles exceeds 30, and the number of bicycles exceeds 30.
 B. The number of tricycles exceeds 30, and the number of bicycles is less than the number of tricycles.
 C. The number of tricycles is less than 100, and the number of bicycles is less than 100.
 D. The number of tricycles and bicycles together exceeds 100, and the number of bicycles is less than 100.
 E. The number of tricycles and bicycles together is less than 100, and the number of tricycles exceeds 30.
58. The horizontal line in the graph below shows the distance between a fixed point and an object as a function of elapsed time. The object could be doing which of the following?

- I. Moving in a circle around the fixed point
 II. Moving in a straight line away from the fixed point
 III. Standing still



- F. I only
 G. II only
 H. III only
 J. I and II only
 K. I and III only



59. In trigonometry, an angle of $-\frac{7\pi}{6}$ radians has the same sine and cosine as an angle that has which of the following degree measures?

A. 30°
B. 60°
C. 120°
D. 150°
E. 210°

60. Which of the following is an equation of the circle with center at $(-2,3)$ and a radius of 5 coordinate units in the standard (x,y) coordinate plane?

F. $x^2 + y^2 + 4x - 6y = 5$
G. $x^2 + y^2 + 4x - 6y = 12$
H. $x^2 + y^2 + 4x - 6y = 25$
J. $x^2 + y^2 - 4x + 6y = 12$
K. $x^2 + y^2 - 4x + 6y = 25$

DO YOUR FIGURING HERE.

END OF TEST 2

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO THE PREVIOUS TEST.

READING TEST

35 Minutes—40 Questions

DIRECTIONS: There are four passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

Passage I

PROSE FICTION: This passage is adapted from the novel *Night Water* by Helen Elaine Lee (©1996 by Helen Elaine Lee).

There had been no words for naming when she was born. She was "Girl Owens" on the stamped paper that certified her birth, and at home, she had just been "Sister," that was all. When asked to decide, at six, what she would be called, she had chosen "Sunday," the time of voices, lifted in praise.

That was one piece of the story, but other parts had gone unspoken, and some had been buried, but were not at rest. She was headed back to claim them, as she had taken her name.

She could smell the burnt, sweet odor of the paper mill that sprawled across the edge of town, and as the train got closer, she remembered all that she saw. She felt herself entering the greens and reds and browns of her own paintings, pulling aside her brushstrokes as if they were curtains and stepping through. There were autumn trees on fire everywhere, and she moved beyond the surface of color and texture into the hidden layers of the past, from which she had learned to speak her life with paint.

The train passed through the part of town where she grew up. She watched as they left behind the neat, compact frame houses and hollow storage buildings. She was going back to piece together their family story of departure and return. She saw it all from the inside out, as native and exile, woman and child. From all that she remembered and all that she was. She was Girl Owens and Sister. She was Sunday, and she was headed home.

Waiting for Sunday's arrival, Delta Owens stepped out onto the front porch. She hoped she would be able to find the right way to approach Sunday, with whom she had only been in touch by mail for five years. She had tried to demonstrate a persistent bond with the help of words put together by experts, choosing for each birthday and holiday an oversized greeting card, depending on its ornate script and polished rhyme to express what she had never been able to say. Each one she had signed "Always, Delta" before addressing the envelope carefully and mailing it off to Chicago. She had had her back to the wall, regularly, receiving wood block prints

or splashes of paint on wefts of heavy paper with ragged edges or on see-through skins. Each one she had turned round and round, looking for right-side up with the help of the signature. Each one she had saved. Though she hadn't known what, specifically, to make of any of them, she knew their appearance said something about the habit of love.

They had kept up contact despite the differences that had accumulated over the years and finally erupted in accusations and insults after Nana's death. In the wide, post-funeral quiet, after the visitors had gone home, they had both uttered things huge and unerasable.

She had always known how Sunday felt about home. "I'm in a little box," she had often complained while growing up, trying to express to Delta how different she felt, how she was of it, but would never be able to stay. And Delta, who had fought anyone who criticized her sister, had listened and comforted her, but hadn't really understood. Sunday was the one she was different from.

"This place pulls you down and holds you," Sunday had said. "Delta, don't you see, it pulls you down and holds you, silent and safe."

What Sunday said that night was condemnation of a place, but Delta absorbed it all. She was of Wake County and caught in that understanding of herself. Intoxicated with saying what had long been felt, they both spoke freely and all barriers fell. Most of the things Sunday said had not surprised Delta, but one indictment had left her open-mouthed: "You don't even see my painting," Sunday had accused, "you don't even see me at all."

Delta had laughed callously at the accusation, for she knew, though she couldn't have said it, that for most of her life she had seen little else. She had answered by calling her a misfit who thought she was better than the folks she had left behind. And it was Delta's recognition of her own rancor, as much as the substance of what they said, that staggered and disgraced her. She hadn't even realized all the things for which she couldn't forgive Sunday, hadn't known her own smallness until she found herself measuring her sister out loud.

Finally, the rush of words had ended, and they had silently straightened up and gone upstairs without repairing their trespasses. Sunday had gathered and packed her things in a wild, tearful stupor of regret and relief, while Delta cried herself to sleep with bitter remorse.

Delta pushed that night from her mind, hoping that this visit might help them leave behind their troubled history.

1. Which of the following statements offers the best short summary of this passage?
 - A. A painter, Sunday, returns to her hometown to attend Nana's funeral and to visit her sister Delta, whom she hasn't seen for some time.
 - B. Two sisters try unsuccessfully to resolve their differences when one of them returns to her hometown for a visit.
 - C. Two sisters contemplate their relationship to each other and to their hometown just prior to a reunion between them.
 - D. Two sisters argue over their childhood differences after one of them considers moving from Chicago back to Wake County.
2. The main conflict in this passage could best be described as:
 - F. Delta's struggle over her inability to express herself adequately in her paintings.
 - G. Sunday's internal conflict over whether she should stay away from or go back to her hometown.
 - H. the tension between two sisters, Delta and Sunday, who have differing views of the world.
 - J. the hostility between two sisters, Delta and Sunday, each of whom wants control over the other.
3. In this passage, the reader gains the most information about how Sunday feels about Wake County from:
 - A. Sunday's words as remembered by Delta.
 - B. Sunday's thoughts.
 - C. the words Delta speaks to Sunday.
 - D. Delta's description of Sunday's artwork.
4. The point of view from which this story is told changes at the beginning of line:
 - F. 21.
 - G. 30.
 - H. 55.
 - J. 66.
5. In the passage, Sunday defines herself as a painter and someone who has broken free of childhood constraints, while Delta tends to define herself as:
 - A. a homemaker, caregiver, and skilled writer in competition with her artistic sister.
 - B. a person who differs from the other residents of Wake County.
 - C. a woman who has always differed from her sister.
 - D. someone who is trying to become more worldly.
6. The first paragraph reveals all of the following details EXCEPT:
 - F. who suggested that Sunday choose her own name.
 - G. what name had been printed on Sunday's birth certificate.
 - H. what the name Sunday represented to Sunday herself.
 - J. at what point "Sister" acquired the name Sunday.
7. As it is used in line 26, the phrase *native and exile* most directly refers to:
 - A. the point of view of an artist versus that of her sister.
 - B. Sunday's love for and conflict with her sister Delta.
 - C. the period of time when Sunday was called "Sister" and the period of time after she became Sunday.
 - D. Sunday's childhood in Wake County and her adulthood in Chicago.
8. According to the passage, how did Delta interpret the paintings on paper and the block prints that Sunday sent to her?
 - F. She believed they indicated Sunday's lack of skill as a painter.
 - G. She thought they were Sunday's way of showing that she cared about her sister.
 - H. She felt they represented Sunday's acceptance that Delta would never leave Wake County.
 - J. She believed they were proof that Sunday was a misfit.
9. According to the passage, Delta had once accused Sunday of:
 - A. not appreciating Delta's artistic talents.
 - B. thinking Sunday was better than the people who lived in Wake County.
 - C. disgracing the family when she moved away from Wake County.
 - D. creating paintings that had almost no significance in the real world.
10. In the context of lines 79–85, the word *smallness* (line 84) most nearly means:
 - F. pettiness.
 - G. willfulness.
 - H. diminutive size.
 - J. lack of artistic talent.

Passage II

SOCIAL SCIENCE: This passage is adapted from Michael J. Sandel's article "America's Search for a New Public Philosophy" (©1996 by The Atlantic Monthly Company). The author argues that our current conception of democracy underlies many of our social and economic problems.

The central idea of the public philosophy by which we live is that freedom consists in our capacity to choose our ends for ourselves. Politics should not try to form the character or cultivate the virtue of its citizens, for to do so would be to "legislate morality." Government should not affirm, through its policies or laws, any particular conception of the good life; instead it should provide a neutral framework of rights within which people can choose their own values and ends.

The aspiration to neutrality finds prominent expression in our politics and law. Although it derives from the liberal tradition of political thought, its province is not limited to those known as liberals, rather than conservatives, in American politics; it can be found across the political spectrum. Liberals invoke the ideal of neutrality when opposing school prayer or restrictions on abortion or attempts by certain groups to bring their morality into the public square. Conservatives appeal to neutrality when opposing attempts by government to impose certain moral restraints—for the sake of workers' safety or environmental protection or distributive justice—on the operation of the market economy.

The ideal of free choice also figures on both sides of the debate over the welfare state. Republicans have long complained that taxing the rich to pay for welfare programs for the poor is a form of coerced charity that violates people's freedom to choose what to do with their own money. Democrats have long replied that government must assure all citizens a decent level of income, housing, education, and health care, on the grounds that those who are crushed by economic necessity are not truly free to exercise choice in other domains. Despite their disagreement about how government should act to respect individual choice, both sides assume that freedom consists in the capacity of people to choose their own ends.

So familiar is this vision of freedom that it might seem a permanent feature of the American political tradition. But as a reigning public philosophy, it is a recent arrival, a development of the past half century. Its distinctive character can best be seen by comparison with a rival public philosophy that it gradually displaced: a version of republican political theory.

Central to republican theory is the idea that liberty depends on sharing in self-government. This idea is not by itself inconsistent with liberal freedom. Participating in politics can be one among the ways in which people choose to pursue their individual ends. According to republican political theory, however, sharing in self-rule involves something more. It involves deliberating with fellow citizens about the common good and

helping to shape the destiny of the political community. But to deliberate well about the common good requires more than the capacity to choose one's ends and to respect others' rights to do the same. It requires a knowledge of public affairs and also a sense of belonging, a concern for the whole, a moral bond with the community whose fate is at stake. To share in self-rule therefore requires that citizens possess, or come to acquire, certain civic virtues. But this means that republican politics cannot be neutral toward the values and ends its citizens espouse. The republican conception of freedom, unlike the liberal conception, requires a formative politics, a politics that cultivates in citizens the qualities of character that self-government requires.

Both the liberal and the republican understandings of freedom have been present throughout our political experience, but in shifting measure and relative importance. In recent decades the civic, or formative, aspect of our politics has given way to a procedural republic, concerned less with cultivating virtue than with enabling persons to choose their own values. This shift sheds light on our present discontent. For despite its appeal, the liberal vision of freedom lacks the civic resources to sustain self-government. The public philosophy by which we live cannot secure the liberty it promises, because it cannot inspire the sense of community and civic engagement that liberty requires.

11. Which of the following offers the best description of republican political theory as it is defined in this passage?
 - A. Citizens should be able to choose their own values, and should respect others' right to do the same.
 - B. Citizens need to participate in politics and to care for their community in order to enjoy true liberty.
 - C. Citizens need to be neutral about their values and beliefs in order that others not be offended.
 - D. Citizens must understand that choosing their own values is inconsistent with liberal freedom.
12. According to the passage, in order for citizens to "deliberate well about the common good" (line 54) citizens must possess all of the following EXCEPT:
 - F. knowledge of public affairs.
 - G. a neutral framework of rights.
 - H. concern for fellow citizens.
 - J. a sense of civic engagement.

13. The main point of the second paragraph (lines 10–23) is that:
- A. liberals and conservatives have essentially the same value systems.
 - B. liberals and conservatives find governmental neutrality appealing for different reasons.
 - C. conservatives are more interested in the nation's economy than are liberals.
 - D. liberals are responsible for the idea that the U.S. government should be neutral.
14. The main point of the fifth paragraph (lines 45–66) is that:
- F. liberal and republican theories each demand that citizens take part in politics.
 - G. liberal and republican theories each have their good points and their bad points.
 - H. republican politicians tend to be more virtuous and compassionate than liberal politicians.
 - J. republican theory requires that citizens demonstrate a concern for the common good.
15. As it is used in line 65, the phrase *formative politics* refers to a type of politics that:
- A. promotes important civic virtues.
 - B. is consistent with liberal freedom.
 - C. is part of a procedural republic.
 - D. is distinct from self-rule.
16. If the last paragraph were deleted, what critical piece of information would be lost?
- F. What the author sees as the features that distinguish between procedural and liberal republics
 - G. What the author sees as the centerpiece of republican political theory
 - H. What the author sees as the central weakness of our current public philosophy
 - J. What the author sees as the difference between liberal and republican political theories
17. As it is used in lines 68–69, the phrase *political experience* most nearly means:
- A. country's history.
 - B. voting patterns.
 - C. presidential campaigns.
 - D. elective office.
18. The main point being made in lines 10–15 is that:
- F. a belief in citizens' rights to choose their own values is shared by people with different politics.
 - G. liberals and conservatives can be found across the political spectrum.
 - H. the aspiration to neutrality is guaranteed by the Constitution and supported by republican theory.
 - J. the liberal tradition in American politics is somewhat limited in its theoretical background.
19. The last paragraph most strongly suggests that a "procedural republic" is a republic:
- A. known for having good political procedures.
 - B. that has developed a formative politics.
 - C. based upon many rules and regulations.
 - D. with a neutral framework of rights.
20. The author of the passage claims that the liberal vision of freedom cannot provide liberty because that vision:
- F. gives citizens too many choices to consider.
 - G. does not teach citizens to care about community.
 - H. inspires moral neutrality, which is anti-government.
 - J. depends too heavily on the concept of legislating morality.

Passage III

HUMANITIES: This passage is adapted from *Bowman's Store: A Journey to Myself* by Joseph Bruchac (©1997 by Joseph Bruchac). The author and his grandfather, Jesse Bowman, are Abenaki Indians.

I remember my grandfather's garden. It was around back, between my grandparents' house and the small two-room building we used for storing things.

Another of my earliest memories is walking through the furrows of that newly plowed garden with my grandfather—who seemed taller than the biggest trees then—holding my hand. I could barely walk, even on level ground, but he wouldn't let me fall. I wore the same kind of overalls that he did, and people were already calling me "Jess's Shadow."

That was how Lawrence Older put it one day when he was buying gasoline at my grandparents' filling station. "Jess," Larry said, with a twinkle in his eye, "that grandson of yours stays so close to you he don't hardly leave room enough for your shadow."

Jess's Shadow? I didn't quite understand that name. My name was Sonny. I knew that for sure. That was the name my grandparents called me by. I never heard either of them speak to me or about me by any other name.

I remember the day—I could not have been more than two and a half years old—when my grandfather said to me, "Sonny, cup out yer bands." I held them out together, trying to make a really good cup. Carefully, taking the seeds out from the cloth sack he had slung over his shoulder, he filled my hands with kernels of golden corn. I stood there holding those seeds for him, watching as he took them, four at a time, to plant them.

"Yer turn now," he said when those seeds were gone, and he held out one leathery hand filled with corn. I'd watched really carefully, so I knew what to do. I did it so well that my grandfather allowed that I was already better at it than he was. Then, as I planted my first hills of corn, he talked to me about things. He always talked more when he was in his garden.

There in his garden, as he spaded or began to hoe, he would talk about the different plants, about the birds we heard singing, about how we had to watch for the woodchucks or the rabbits or the raccoons. Sometimes he would talk about the old way of plowing with a team of horses, and he would tell me the names of the horses he'd loved—the last of them dead and gone a decade before I was born. He told me how, as their plows cut their way down into the rich earth that was the same color as his face, he and his father used to turn up arrowheads.

"Indian arrowheads," he would say.

He had a name for that work of preparing his garden for the sweet corn and green beans and butternut squash we always put in. I never heard anyone else say

it just the way he did until I met Grampa's younger brother Jack, many years later, when I was a grown man. "Fitting the ground to plant." That is what Grampa called it. This is what Jack called it. This is what their father, Lewis Bowman, had called it. Fitting the ground to plant.

Another spring day, perhaps a year later, I was following my grandfather while he worked in the garden. He had made a small hoe for me that year, and I was, as always, trying to do exactly what he did. Then he stopped hoeing. I stopped too and waited. I remember that I heard a bird sing just then, a long ululating song.

"Oriole," my grandfather said. Then he bent over and picked something up. He brushed soil from it and went down on one knee next to me.

I leaned close to look at the dark, blocky piece of flint that filled my grandfather's broad hand.

"Indian," my grandfather said. "Axhead."

He hefted it first in one hand and then the other. He did it with the same care that my grandmother used when she was gathering eggs in the henhouse and putting them into the basket hung over her arm. Then he placed that axhead into my hands.

Thirty years after that day when my grandfather put the stone axhead in my hands, I understood at last why, as I held that stone, my mind had filled with images of tall corn swaying in the wind, images of women dancing as they held the season's harvest in their hands. Hearing a friend tell the story of the coming of corn was the last stroke of the hoe that fit my own mind to the earth my grandfather had given to me. And I knew that as long as my hands had the strength to hold a hoe, I would work that garden where corn had been cared for by my grandparents and by my great-grandparents before them. I would listen to that land just as it had once been listened to by other men and women, generations of Abenaki people and Mohawk people whose stories were told in a tongue as old as the soil. Spring would find me preparing the earth for the Corn Maiden, find me fitting the ground of my grandfather's garden to plant.

21. Which of the following statements best describes the nature of this passage?

- A. A grandson learns to appreciate his grandfather, whom he had always had difficulty understanding before.
- B. An adult reflects on time spent with his grandfather and comes to understand more fully what he learned from him.
- C. A grown man looks back with nostalgia on his relationship with his grandfather and on a garden that no longer exists.
- D. A grandson recounts childhood memories that bring back strong and often unpleasant emotions from his years spent with his grandfather.

22. The perspective from which the narrator relates the passage shifts most dramatically beginning with line:
- F. 16.
 - G. 21.
 - H. 57.
 - J. 74.
23. It can reasonably be inferred that within the passage, the nickname Jess's Shadow represents all of the following about the narrator EXCEPT his:
- A. strong emotional bond with his grandfather.
 - B. close physical proximity to his grandfather when the narrator was young.
 - C. conscious imitation of his grandfather.
 - D. feeling of being protected from other people by his grandfather.
24. In the fifth and sixth paragraphs (lines 21–35), the narrator approaches planting corn with his grandfather with what is best characterized as:
- F. a reluctant willingness to participate in an activity the narrator knows is important to his grandfather.
 - G. a great enthusiasm that causes the narrator to be slightly careless in his work.
 - H. an interest in every detail of the process in an effort to do the best job possible.
 - J. a genuine interest in the work that fades as the work becomes more difficult.
25. The scene revealed in the tenth through fourteenth paragraphs (lines 57–73) is best described by which of the following?
- A. An older man shares with his grandson a symbol of their common heritage.
 - B. Two people are brought closer together by their love of birds.
 - C. A young child comes to value a small hoe given to him by his grandfather.
 - D. The discovery of an Indian axhead leads a grandfather to a new appreciation of his past.
26. The images in lines 4–8 are used primarily to convey the narrator's sense as a young child of being:
- F. overly controlled by his grandfather.
 - G. watched over by his awe-inspiring grandfather.
 - H. puzzled by his grandfather's height.
 - J. impressed by the size of his grandfather's garden.
27. The narrator suggests that Lawrence Older's comment, quoted in lines 13–15, was intended to express:
- A. amusement.
 - B. concern.
 - C. dismay.
 - D. elation.
28. According to the passage, the narrator initially reacts to being called Jess's Shadow with confusion because:
- F. his parents had given him the name Sonny.
 - G. he is no longer certain whether his name is Sonny or Jess's Shadow.
 - H. he is unsure whether Lawrence Older was referring to him or to someone else.
 - J. Sonny is the only name he remembers his grandparents using for him.
29. The narrator describes the term "fitting the ground to plant" as one that was:
- A. commonly used by people in the community where his grandfather lived.
 - B. invented by his grandfather and used by the narrator.
 - C. traditionally used by people in his grandfather's family.
 - D. unfamiliar to the narrator until he was a grown man.
30. The author includes the fourteenth paragraph (lines 69–73) primarily to:
- F. introduce the character of the grandmother, who up to this point has been absent from the narrative.
 - G. describe the type of work that the narrator's grandparents often had to do.
 - H. establish that the narrator and his grandfather would often exchange small but meaningful gifts.
 - J. portray the reverence with which the grandfather treated the object he had unearthed.

Passage IV

NATURAL SCIENCE: This passage is adapted from the book *Eggs: Nature's Perfect Package* by Robert Burton (©1987 by Robert Burton).

Birds are 'glorified reptiles' and some biologists believe that they are indeed no more than reptiles with feathers. It is not surprising, therefore, that birds' eggs are similar to those of reptiles. The main interest in birds' eggs from an evolutionary point of view is the adaptation of the eggs of different species to particular lifestyles.

In both reptile and bird eggs, there is a large yolk which not only supplies the needs of the embryo but will also sustain the newly hatched animal until it can feed. In birds' eggs, after the egg cell—consisting mainly of a ball of yolk—has been fertilized, it moves down the oviduct where it collects the albumen or 'egg white' and takes on its shell. The albumen is mainly a filler for protecting the embryo, but it also supplements the yolk as a valuable source of protein and a reservoir of water. Between the albumen and the shell are two parchment-like shell membranes. When they are first formed they fit quite loosely, but during the next stage, known as 'plumping', water and salts fill out the albumen until the membranes are taut. The shell is then attached firmly onto the outer membrane, as can be seen when peeling a hard-boiled egg.

The innermost layer of albumen is thick and forms a hammock slung between the ends of the shell to keep the yolk suspended in the centre. Within it are two twisted cords called chalazae, which act as bearings to allow the yolk and embryo to rotate freely when the egg is turned by the parent bird in the nest. Chalazae are presumably unnecessary in reptile eggs because they are never turned, and some, such as those of tortoises, will not hatch if they are turned accidentally.

The shell takes 15 to 16 hours to form in a domestic hen. It consists of crystals of calcite, a form of calcium carbonate, strengthened with protein fibres which also attach it to the underlying membrane. The amount of calcium needed for each egg is about 2 grams. As no more than 25 milligrams are circulating in the blood of the hen, the balance must be mobilized from her reserves. She cannot absorb enough calcium from her food while making the egg, so she has to remove it from her bones. Birds have developed a special kind of bone, called medullary bone, which females lay down in the marrow cavity of existing bones during the weeks before laying commences and use as a source of calcium when the eggshells are being formed. Some birds even have difficulty in laying down this reserve because their diet is deficient in calcium. Shortage of calcium affects the breeding of scavenging birds, such as vultures, which have a diet of pure meat; to overcome this problem some species switch to hunting small animals, which they swallow whole, during the breeding season. Another example of the ingenious methods used by birds to provide calcium for their eggs

can be seen on the Arctic tundra, where insect-eating sandpipers swallow the bones of long-dead lemmings; as a result there is more calcium in each clutch of eggs than there is in the adult's entire skeleton.

The shell of a hen's egg is peppered with 10,000 tiny pores, just visible to the naked eye as small depressions in the shell. These pores are used for the passage of oxygen, carbon dioxide and water vapour. Unlike a lung, however, the flow of gases cannot be regulated and relies purely on diffusion through the pores. Nevertheless, the flow must be finely attuned to the needs of the embryo. As development proceeds, more oxygen is required, but the loss of water must be kept low. The eggshell's role in regulating gas flow and water loss is shown by the arrangement of pores in different eggs—quite simply, diffusion depends on the size and number of the pores. Of all the eggs studied, from the 1-gram eggs of warblers to the 1.5-kilogram ostrich eggs, the weight of the egg and hence its metabolic requirements is directly proportional to the total cross-sectional area of the pores.

Birds' eggs are uniform in shape when compared with the eggs of reptiles, but there are significant variations from the standard oval. Owls' eggs are noticeably more spherical than normal, while those of swifts and swallows are longer and more elliptical, although it is not obvious why this should be. Many waders have eggs shaped like old-fashioned spinning tops, blunt at one end and tapering sharply at the other. The reason for this is clear: there are usually four in a clutch and, being large for the size of the parent, they fit under its body better when packed points inward in the nest.

31. Which of the following best describes the content of the passage?
 - A. A point by point comparison between the eggs of birds and those of reptiles
 - B. A technical discussion of the development of birds' eggs, including some references for comparison sake to those of reptiles
 - C. An examination of how recent studies have shown that birds' eggs are less like reptile eggs than was previously thought
 - D. A discussion of how from an evolutionary standpoint birds' eggs have become more adapted to their environment than reptile eggs have
32. Which of the following statements about the eggshell of a domestic hen is NOT supported by the passage?
 - F. It contains about 10,000 tiny pores, which are visible to the naked eye.
 - G. It takes about 15 hours to form.
 - H. It consists of crystals of calcite.
 - J. It contains about 15 grams of calcium.

33. Which of the following statements best explains how the last paragraph functions in the passage as a whole?
- A. It summarizes the main points made about eggs in the preceding paragraphs.
 - B. It introduces a comparison between reptile eggs and bird eggs.
 - C. It brings the issue of shape into the discussion of significant features of an egg.
 - D. It focuses attention on what puzzles biologists most about the design and function of eggs.
34. What explanation, if any, does the passage provide for why a parent bird turns the eggs in the nest?
- F. In some species, the parent turns the egg to maximize the exposure of the pores to oxygen and water.
 - G. The parent bird turns the egg to ensure the equal distribution of its body heat to the entire clutch.
 - H. The parent bird turns the egg as one of many measures it takes to determine if the egg is developing properly.
 - J. The passage provides no explanation for this behavior.
35. The passage indicates that from the inside out a bird's egg is made up of the following elements in the following order:
- A. yolk, chalazae-containing albumen, two shell membranes, shell.
 - B. yolk, egg cell, inner shell, parchment membranes, outer shell.
 - C. yolk, parchment membrane, cell membrane, albumen, inner shell, chalazae, outer shell.
 - D. yolk, chalazae, parchment membranes, albumen, shell.
36. As it is used in line 20, the word *plumping* refers to the:
- F. phase in which the adult female bird increases her nutrient intake in preparation for egg-laying.
 - G. formation of two parchment-like membranes around the fertilized egg cell.
 - H. stage during which increased levels of water and salts expand the albumen in a developing egg.
 - J. adjustments a nesting bird makes in its plumage to maximize its ability to warm a clutch of eggs.
37. Which of the following best describes the form and function of the chalazae as they are described in the passage?
- A. A hammock inside the shells of birds and reptiles that protects the developing embryo
 - B. Two twisted cords that allow the yolk and embryo in a bird's egg to rotate freely when the egg is moved
 - C. Two small vessels that allow the exchange of nutrients and waste between the yolk and the embryo in a bird's egg
 - D. Two thin layers of protein that protect the embryo in bird and reptile eggs once plumping has occurred
38. According to the passage, protein fibers in the eggshells of birds serve to:
- F. deliver crystals of calcite to the bones of the developing embryo.
 - G. facilitate the movement of life-sustaining gases to the embryo.
 - H. strengthen the shell and attach it to the underlying membrane.
 - J. suspend the yolk in a hammock-like feature for protective purposes.
39. Which of the following best describes the medullary bone and its significance in bird development?
- A. It is the lower of two hollow bones that together form the legs of a bird.
 - B. It is a calcium-rich bone in small Arctic mammals that become the preferred prey of breeding Arctic birds.
 - C. It is a pliable bone that temporarily encases the spinal cord in developing bird and reptile embryos until it is replaced by a stronger bone.
 - D. It is a supplemental bone that forms inside a hen's existing bones prior to egg-laying to serve as a source of calcium.
40. What advantage, if any, does the passage suggest is offered by the shape of the eggs of waders?
- F. The shape provides protection from certain kinds of egg-eating predators that don't recognize the wader egg as an egg.
 - G. The shape allows the egg to be positioned efficiently along with other eggs underneath the nesting adult wader.
 - H. The shape provides the embryo the space it needs to extend and thereby strengthen its developing wings.
 - J. The passage does not indicate what advantage the shape of these eggs provides.

END OF TEST 3

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO A PREVIOUS TEST.

SCIENCE TEST

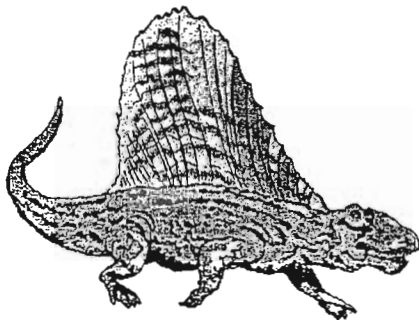
35 Minutes—40 Questions

DIRECTIONS: There are seven passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

You are NOT permitted to use a calculator on this test.

Passage 1

Dimetrodon is a large, extinct, cold-blooded, carnivorous reptile (see figure below). Fossils show that *Dimetrodon* had fixed spines extending from its vertebrae that, when covered with skin, formed a large sail. Two scientists discuss the sail's function.



Scientist 2

The sail was used for display. First, the sail made the reptile appear larger, thereby discouraging attacks from predators. Second, sails were smaller in female *Dimetrodons* than in males, indicating the sails were used by the males for attracting mates. Modern lizards change their skin color and pattern to attract a mate. Similarly, *Dimetrodons* controlled the appearance of the skin on the sail by regulating the blood supply to the sail.

Sphenacodon is a cold-blooded reptile that lived at the same time and in the same climate as *Dimetrodon*. *Sphenacodon* is a close relative of *Dimetrodon* and is identical in all details of its anatomy, except that it did not have a sail or any other projections on its back. If sails were needed for temperature regulation, *Sphenacodon* would have had one also.

Mathematical calculations show that as a *Dimetrodon* sail absorbed heat on the side facing the sun, it may have lost nearly as much heat from the side facing away from the sun.

Scientist 1

The sail helped regulate body temperature. *Dimetrodon* needed to warm its body to the *activity temperature*, the temperature at which it was active enough to hunt. This could be achieved most rapidly by turning the broad side of the sail toward the morning sun. A mathematical model shows that a reptile with a sail would reach the activity temperature 3 times faster than a reptile without a sail. Rapid warming gave *Dimetrodon* a distinct advantage catching cold-blooded prey.

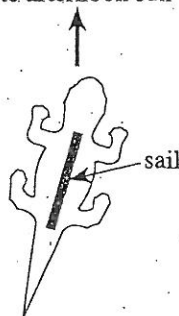
Grooves on the sail's spines held large blood vessels. Regulating the blood flow through these vessels during the heat of the day allowed the reptile to radiate heat away from its body and prevent overheating.

Four other cold-blooded reptile species from this same time and climate also had bony projections or sails on their back. Today, elephants regulate their body temperature by flapping their large ears.

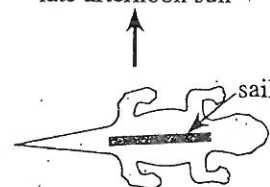
- According to the mathematical calculations mentioned by Scientist 2, a *Dimetrodon* that turned its sail toward the morning sun would experience:
 - rapid overheating and death.
 - little or no increase in body temperature.
 - an increase in the length of the sail spines.
 - a decrease in the blood supply to the sail.
- If Scientist 1 is correct, exposing its sail to a strong wind would have which of the following effects on a *Dimetrodon*'s body temperature? Exposing its sail to a wind at a temperature less than the body temperature of *Dimetrodon* would:
 - warm the sail and lower the body temperature.
 - warm the sail and raise the body temperature.
 - remove heat from the sail and lower the body temperature.
 - remove heat from the sail and raise the body temperature.

3. According to Scientist 2, male *Dimetrodons* would control the blood supply to their sails in order to:
- increase or decrease the spines' width.
 - reduce their body temperature.
 - change the sails' color or pattern.
 - raise or lower their sails.
4. Both scientists would most likely agree with which of the following statements about *Dimetrodons* and modern animals?
- There are similarities between *Dimetrodons* and some modern animals.
 - There are no similarities between *Dimetrodons* and any modern animals.
 - There are no modern animals that are direct or indirect descendants of *Dimetrodons*.
 - There are many modern animals that look exactly like *Dimetrodons*.
5. According to Scientist 1, a cold-blooded, carnivorous reptile without a sail would have compared to a *Dimetrodon* of the same size in which of the following ways? This reptile without a sail would have:
- reached the activity temperature more slowly than did *Dimetrodon*.
 - reached the activity temperature more quickly than did *Dimetrodon*.
 - been much easier to see than was *Dimetrodon*.
 - been less easily preyed upon than was *Dimetrodon*.
6. Both scientists would most likely agree with which of the following statements about *Dimetrodons* and their sails? *Dimetrodons* were able to:
- raise and lower their sails at will.
 - shed and regrow their sails periodically.
 - control the blood supply to their sails.
 - regulate body temperature without their sails.
7. Based on the information in Scientist 1's viewpoint, which of the following figures best represents how a *Dimetrodon* would orient its sail with respect to the sun in order to *minimize* heating during late afternoon?

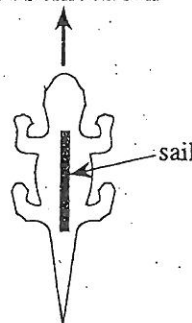
A. direction of late afternoon sun



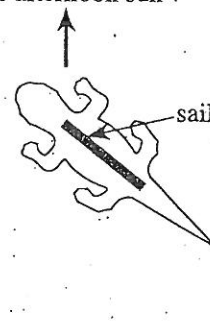
C. direction of late afternoon sun



B. direction of late afternoon sun



D. direction of late afternoon sun



Passage II

The *rate* of a chemical reaction can be measured as the change in the concentration of its products over a period of time. The following experiments were performed to investigate factors that influence the rate at which potassium permanganate (KMnO_4) reacts with oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$) to form Mn^{2+} and other products.

Experiment 1

A 4.0 mL sample of 0.1 M (moles/liter) aqueous KMnO_4 , which was purple in color, was measured with a clean graduated cylinder and poured into a test tube. A 4.0 mL sample of 1.0 M aqueous $\text{H}_2\text{C}_2\text{O}_4$ was then measured with a clean graduated cylinder and poured into a second test tube. The 2 test tubes were placed in a 25°C water bath. After 5 minutes the $\text{H}_2\text{C}_2\text{O}_4$ solution was mixed with the KMnO_4 solution and placed back in the water bath. The time until the purple color disappeared was recorded. This procedure was repeated at 3 other temperatures. The results are shown in Table 1.

Table 1		
Trial	Temperature ($^\circ\text{C}$)	Reaction time (sec)
1	25	210
2	35	105
3	45	48
4	55	25

Experiment 2

The procedure from Experiment 1 was repeated in every way except that 0.1 mg of manganous sulfate (MnSO_4), a *catalyst*, was added to each test tube containing KMnO_4 before the $\text{H}_2\text{C}_2\text{O}_4$ was added. Catalysts are substances that increase the rate of reactions without being used up. The results are shown in Table 2.

Table 2		
Trial	Temperature ($^\circ\text{C}$)	Reaction time (sec)
5	25	66
6	35	43
7	45	20
8	55	12

Experiment 3

The procedure from Experiment 1 was repeated at a constant temperature of 25°C and various concentrations of $\text{H}_2\text{C}_2\text{O}_4$. The results are shown in Table 3.

Table 3		
Trial	Concentration of $\text{H}_2\text{C}_2\text{O}_4$ (M)	Reaction time (sec)
9	0.125	837
10	0.250	625
11	0.500	415

8. In Experiment 1, as the reaction temperature increased, the time until the purple color disappeared:
 - F. remained the same.
 - G. decreased only.
 - H. increased only.
 - J. increased, then decreased.
9. If the procedure described in Experiment 2 was repeated and a reaction time of 6 sec was measured, the temperature of the reaction was most likely:
 - A. less than 25°C .
 - B. greater than 25°C and less than 40°C .
 - C. greater than 40°C and less than 55°C .
 - D. greater than 55°C .
10. How is the experimental design of Experiment 1 different from that of Experiment 2? In Experiment 1:
 - F. MnSO_4 was not added to the reaction; in Experiment 2 MnSO_4 was added.
 - G. the concentration of KMnO_4 was less than that used in Experiment 2.
 - H. reaction temperatures were varied; in Experiment 2 they remained constant.
 - J. the concentration of $\text{H}_2\text{C}_2\text{O}_4$ was varied; in Experiment 2 it remained constant.



11. Each of the following factors was investigated in the experiments EXCEPT:
- A. adding a catalyst to the reaction mixture.
 - B. changing the reaction solvent.
 - C. varying the reaction temperature.
 - D. changing the concentration of a reactant.
12. A student performed the procedure described in Experiment 1, mixing an $\text{H}_2\text{C}_2\text{O}_4$ solution at 45°C with a KMnO_4 solution at 40°C . Based on the results of Experiment 1, the reaction time was most likely:
- F. less than 25 sec.
 - G. greater than 25 sec and less than 48 sec.
 - H. greater than 48 sec and less than 105 sec.
 - J. greater than 105 sec.
13. A student repeating Trial 9 unknowingly used a graduated cylinder that already contained 1.0 mL of H_2O to measure the $\text{H}_2\text{C}_2\text{O}_4$ solution. Based on the results of Experiment 3, the reaction time that she measured was most likely:
- A. less than 415 sec.
 - B. greater than 415 sec and less than or equal to 625 sec.
 - C. greater than 625 sec and less than or equal to 837 sec.
 - D. greater than 837 sec.

Passage III

As a living cell's *surface area-to-volume ratio* (S/V) increases, so does the cell's ability to exchange substances with its environment.

To model this exchange process, different-sized cubes of gelatin were immersed in a dye for 1 minute. Table 1 lists the edge length, surface area, volume, S/V , and percent volume of gelatin that was dyed.

Table 1				
Edge length (mm)	Surface area (mm ²)	Volume (mm ³)	S/V (mm ⁻¹)	Percent volume dyed
1	6	1	6	100
2	24	8	3	100
4	96	64	1.5	84
8	384	512	0.75	43
16	1,536	4,096	0.375	8

Figure 1 shows the S/V of certain rectangular solids and spheres.

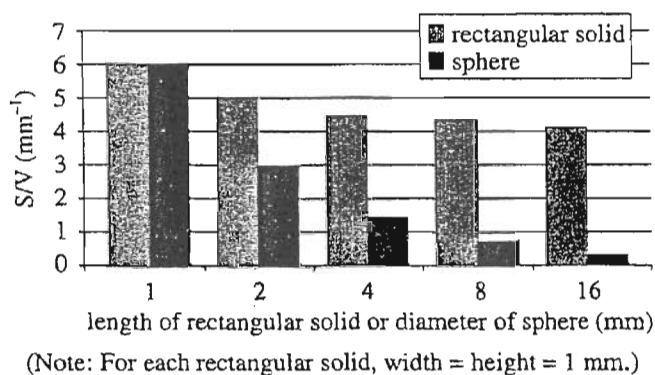


Figure 1

Figure 2 shows the volume of certain rectangular solids and spheres as a function of surface area.

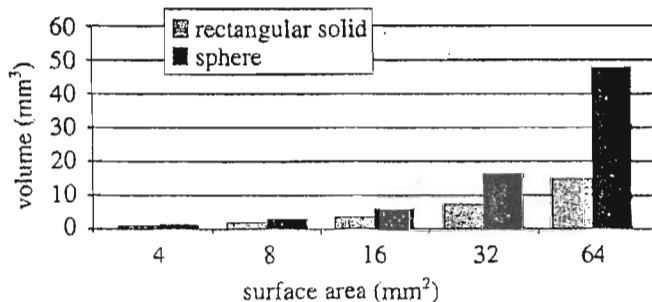


Figure 2

Table 2 lists the diameter and surface area of 5 types of spherical ova (egg cells).

Table 2		
Type	Diameter (mm)	Surface area (mm ²)
Sea urchin	0.07	0.02
Human	0.1	0.03
Hummingbird	6	100
Hen	25	2,000
Ostrich	100	30,000

Table 2 adapted from Norman K. Wessells and Janet L. Hopson, *Biology*. ©1988 by Random House, Inc.

14. Ova with very large volumes tend to contain large amounts of nutrients that are used for growth by the embryo. Based on the information in Table 2 and Figure 2, which of the following ova most likely contain the greatest amount of nutrients?

F. Sea urchin ova
 G. Human ova
 H. Hummingbird ova
 J. Hen ova

15. Based on the information in Table 1, a cube-shaped cell with which of the following edge lengths will have the greatest S/V ?

A. 20 mm
 B. 40 mm
 C. 60 mm
 D. 80 mm



16. Which of the following best describes why the units for the S/V in Figure 1 are mm^{-1} ?
- F. The units for volume are mm and the units for surface area are mm^2 .
 - G. The units for volume are mm^2 and the units for surface area are mm^3 .
 - H. The units for surface area are mm and the units for volume are mm^2 .
 - J. The units for surface area are mm^2 and the units for volume are mm^3 .
17. Based on the information in Figure 1, a spherical cell with a diameter of 6 mm will have an S/V closest to which of the following values?
- A. 0.1 mm^{-1}
 - B. 1 mm^{-1}
 - C. 2 mm^{-1}
 - D. 10 mm^{-1}
18. Based on the information in Table 1, if a cube-shaped plant cell had an edge length of 0.5 mm, it would have an S/V of:
- F. 3 mm^{-1}
 - G. 6 mm^{-1}
 - H. 12 mm^{-1}
 - J. 18 mm^{-1}

Passage IV

Some students were interested in why antifreeze is mixed with the H_2O in a car's radiator. They knew that when a substance is dissolved in H_2O , the boiling points are elevated and the freezing points are lowered relative to H_2O . They performed the following experiments to investigate the effects of the addition of *ethylene glycol* ($\text{C}_2\text{H}_6\text{O}_2$), the active ingredient in most types of antifreeze, to H_2O .

Experiment 1

Six solutions of various concentrations of $\text{C}_2\text{H}_6\text{O}_2$ in H_2O were prepared. The mass of 10.00 mL of each solution was measured, and the density of each solution was calculated using the following formula:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

The densities of 100% H_2O and 100% $\text{C}_2\text{H}_6\text{O}_2$ were also determined. The students found that as the concentration of $\text{C}_2\text{H}_6\text{O}_2$ increased, the density of the solution increased.

Experiment 2

A 20 mL sample of H_2O was placed in a flask containing a thermometer and the flask was heated. The temperature at which the H_2O first started to boil was recorded as the boiling point. This procedure was repeated with $\text{C}_2\text{H}_6\text{O}_2$ and the solutions prepared in Experiment 1 (see Figure 1).

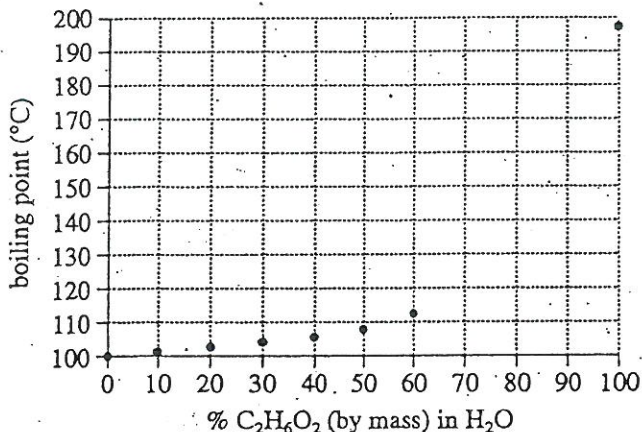


Figure 1

Experiment 3

A 90 g sample of H_2O (freezing point = 0°C) and 10 g of $\text{C}_2\text{H}_6\text{O}_2$ (freezing point = -11.5°C) were placed in a cup containing a thermometer. The mixture was stirred as it was cooled in a low-temperature bath. The temperature at which ice crystals first began to appear was recorded as the freezing point. The procedure was repeated with different amounts of H_2O and $\text{C}_2\text{H}_6\text{O}_2$ (see Table 1).

Mass of $\text{C}_2\text{H}_6\text{O}_2$ (g)	Mass of H_2O (g)	Freezing point ($^{\circ}\text{C}$)
10	90	-3.4
20	80	-7.9
30	70	-14.6
40	60	-23.8
50	50	-36.1
60	40	-51.2

19. If Experiment 2 were repeated with an aqueous solution containing 80% $\text{C}_2\text{H}_6\text{O}_2$ by mass, the boiling point of the solution would most likely be:
 - A. less than 100°C .
 - B. between 100°C and 110°C .
 - C. between 110°C and 115°C .
 - D. greater than 115°C .
20. Where the students live, the lowest temperature during the past year was -31°C . Based on the results of Experiment 3, which of the following percents, by mass, of $\text{C}_2\text{H}_6\text{O}_2$ in H_2O would have been necessary to be sure that the mixtures in car radiators in this area did not freeze?
 - F. 20%
 - G. 30%
 - H. 40%
 - J. 50%

21. Based on the results of Experiment 1, the mass of 10.00 mL of which of the following aqueous solutions would be the greatest?
- A. 10% $\text{C}_2\text{H}_6\text{O}_2$
 - B. 20% $\text{C}_2\text{H}_6\text{O}_2$
 - C. 30% $\text{C}_2\text{H}_6\text{O}_2$
 - D. 40% $\text{C}_2\text{H}_6\text{O}_2$
22. A student repeating Experiment 3 placed an aqueous mixture containing 65% $\text{C}_2\text{H}_6\text{O}_2$ in the low-temperature bath. She did not observe the formation of any crystals. Which of the following statements best explains this observation?
- F. The temperature of the low-temperature bath was exactly the same as the mixture's freezing point.
 - G. The temperature of the low-temperature bath was higher than the mixture's freezing point.
 - H. The temperature of the low-temperature bath was lower than the mixture's freezing point.
 - J. The mixture will not freeze and therefore has no freezing point.
23. Based on the results of Experiment 2, why would it be better to use a mixture of antifreeze and H_2O rather than pure H_2O in a car radiator during hot weather? The mixture has a:
- A. higher boiling point than pure H_2O .
 - B. lower boiling point than pure H_2O .
 - C. higher freezing point than pure H_2O .
 - D. lower freezing point than pure H_2O .
24. Based on the passage and the results of Experiment 2, which of the following types of water would most likely have the *lowest* boiling point?
- F. Lake water
 - G. Distilled water
 - H. Salt water
 - J. Tap water

Passage V

Heat conduction is a process by which heat is transferred from a region at a higher temperature to a region at a lower temperature. Two regions that are in *thermal equilibrium* with each other are at the same temperature. No heat is transferred between regions in thermal equilibrium with each other.

A student studied the relationship between the length and diameter of a copper wire connecting 2 metallic blocks and the rate of heat conduction between the blocks.

Experiment 1

A small aluminum block, A, was heated to 100°C in boiling water, and an identical aluminum block, B, was cooled to 0°C in ice water. Both blocks were immediately placed on an *insulating surface* (a surface that does not conduct heat well) and connected by a given length of 1 mm diameter copper wire initially at room temperature so that the wire would conduct heat from Block A to Block B (see Figure 1).

A:Left B:Right

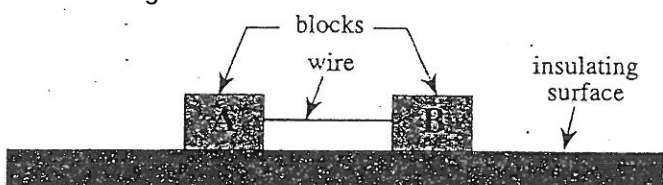


Figure 1

The temperature of each block was measured at regular time intervals for 3 minutes. The same procedure was done with different lengths of 1 mm diameter copper wire. The results are shown in Table 1.

Table 1						
Length of wire (cm)	Temperature (°C) of Block A after:			Temperature (°C) of Block B after:		
	1 min	2 min	3 min	1 min	2 min	3 min
10	79	66	54	18	25	31
20	88	80	72	9	15	19
30	91	85	78	6	10	14

Experiment 2

The procedure described in Experiment 1 was repeated using a copper wire with a diameter of 1.4 mm. The results are shown in Table 2.

Table 2						
Length of wire (cm)	Temperature (°C) of Block A after:			Temperature (°C) of Block B after:		
	1 min	2 min	3 min	1 min	2 min	3 min
10	61	45	40	34	45	40
20	79	65	53	17	25	35
30	85	75	66	12	16	20

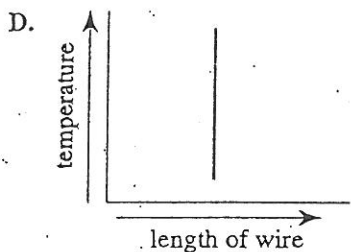
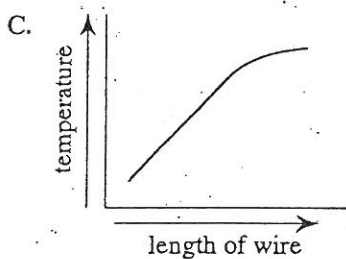
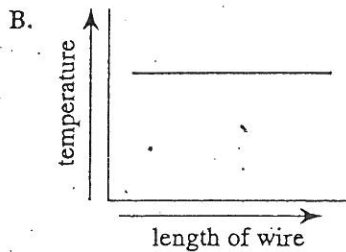
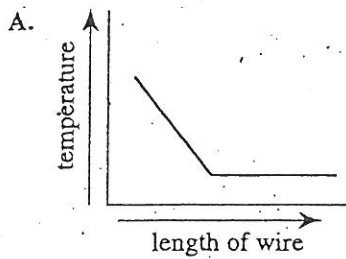
25. The best way to test the hypothesis that the rate of heat conduction depends on the type of wire material used in the experiments would be to repeat the experiments using:

A. silver wires and aluminum wires differing in size from the copper wires.
 B. silver wires and aluminum wires having the same sizes as the copper wires.
 C. copper wires each having a different diameter.
 D. copper wires each having a different length.

26. During Experiments 1 and 2, the 2 blocks reached thermal equilibrium when which of the following copper wires was used?

F. The 10 cm long, 1 mm diameter wire
 G. The 30 cm long, 1 mm diameter wire
 H. The 10 cm long, 1.4 mm diameter wire
 J. The 30 cm long, 1.4 mm diameter wire

27. Based on Experiments 1 and 2, which of the following graphs best represents the relationship between the length of a copper wire of a given diameter and the temperature of Block A at a given time?



28. As *thermal conductivity* decreases, the ability to conduct heat decreases. Brass has a thermal conductivity one-third that of copper. If, in any of the experiments, a copper wire were replaced with a brass wire of the same length and diameter, which of the following outcomes would be most likely?

- F. Block A would lose heat more slowly than when a copper wire was used.
- G. Block A would lose heat at the same rate as when a copper wire was used.
- H. Block B would gain heat at the same rate as when a copper wire was used.
- J. Block B would gain heat more rapidly than when a copper wire was used.

29. In each of the 2 experiments, the amount of heat lost by Block A was greater than the amount of heat gained by Block B. The most probable explanation for this inequality is that:

- A. Block A lost heat to the air.
- B. the copper wire lost heat to Block A.
- C. Block B lost heat to the copper wire.
- D. the air lost heat to Block B.

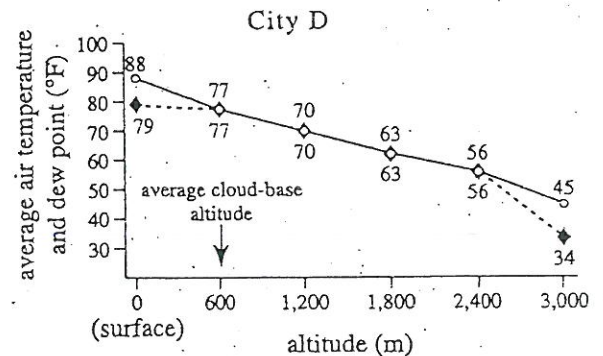
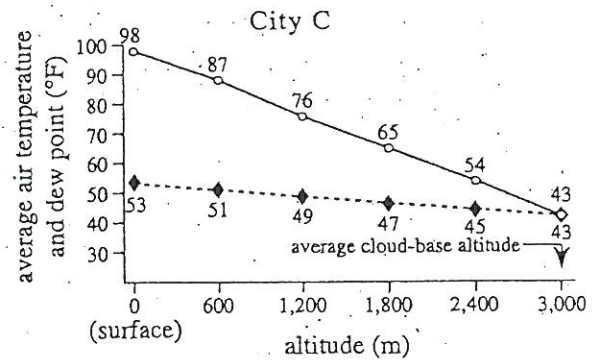
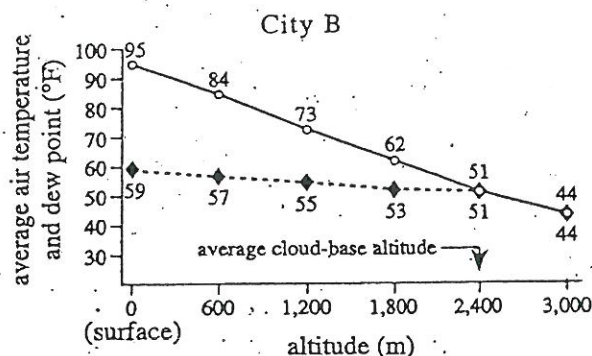
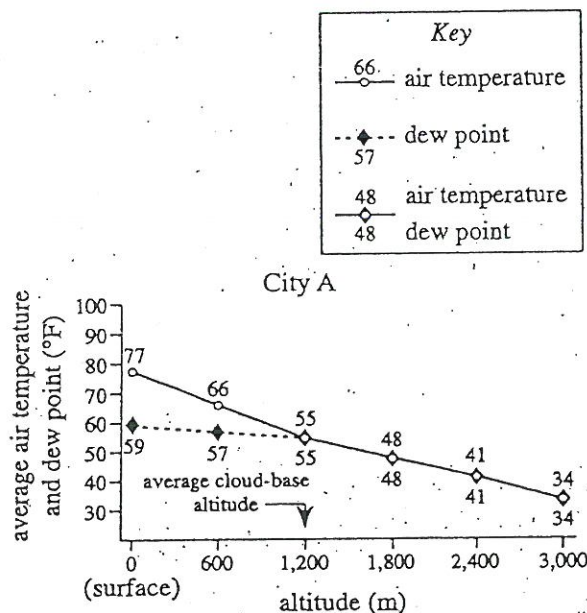
30. Suppose, after 4 minutes, the temperature of Block A was 64°C and that of Block B was 23°C . According to the results of the experiments, which of the following copper wires was most likely being used to connect the blocks?

- F. A 1 mm diameter wire, 10 cm long
- G. A 1 mm diameter wire, 20 cm long
- H. A 1.4 mm diameter wire, 10 cm long
- J. A 1.4 mm diameter wire, 20 cm long

Passage VI

Air at its *dew point temperature* (or *dew point*) is saturated with water vapor. Air temperature usually decreases with increasing altitude. When air cools to its dew point, the water vapor in the air can begin to condense and form clouds.

Air temperatures and dew points were measured every 600 m from the surface (0 m) to an altitude of 3,000 m at 4 cities (A–D). Data, including the altitude of the cloud base over each city, were taken at 2 P.M. on every summer day that clouds were present. Figures 1–4 show the average air temperatures, the average dew points, and the average cloud-base altitudes at each city.



31. A pilot flying a plane at 1,200 m altitude notices that there are cloud layers both above and below her, but none at her current altitude. Which of the following statements about the weather conditions at or near her altitude is supported by the information provided?

- A. The base of the upper cloud layer is at 600 m altitude and extends 1,200 m upward.
- B. The top of the lower cloud layer is at 1,202 m altitude.
- C. The air temperature is higher than the dew point at 1,200 m altitude.
- D. The dew point is above 80°F in both the upper and the lower cloud layers.

32. A *ceilometer* bounces a light beam off the base of a cloud layer and measures how long it takes the light to make the trip up and back. The scientists who made the measurements plotted in Figures 1–4 most likely used the ceilometer to determine which of the following data?

- F. The average air temperature at the top of the cloud layers at Cities A–D
- G. The average dew point of the air at the top of the cloud layers at Cities A–D
- H. The average altitude of the bottom of the cloud layers at Cities A–D
- J. The average thickness of the cloud layers at Cities A–D

33. Assume that the cloud layer at City C extends from an altitude of 3,000 m up to an altitude of 4,000 m. Based on this assumption and the observed trends within cloud layers for the 4 cities, which of the following would be the best approximation of the air temperature and dew point in the cloud layer at an altitude of 3,600 m over City C?

	Air temperature	Dew point
A.	26°F	36°F
B.	36°F	36°F
C.	38°F	45°F
D.	45°F	45°F

34. The table below shows the average July air temperatures and dew points at 0 m altitude for Miami, Florida, and for Phoenix, Arizona.

City	Average air temperature (°F)	Average dew point (°F)
Miami	85	70
Phoenix	95	36

Based on this information and the data in Figures 1–4, which of these 2 cities, on average, would have lower cloud bases in July?

- F. Phoenix, because it has a higher average air temperature than Miami.
- G. Phoenix, because the difference between the average air temperature and the average dew point is larger than in Miami.
- H. Miami, because it has a lower average air temperature than Phoenix.
- J. Miami, because the difference between the average air temperature and the average dew point is smaller than in Phoenix.

35. A certain city reported that the air temperature and the dew point at 0 m altitude measured at 8 A.M. on a given day were both 40°F. Given this information and the data provided, one could be most certain that clouds were present at which of the following altitudes at that city at 8 A.M. on that day?

- A. 0 m
- B. 600 m
- C. 1,200 m
- D. 2,400 m

Passage VII.

The amount of O_2 present in blood and tissues can be measured, in mm Hg, as the *partial pressure of O_2* (P_{O_2}). The *percent saturation of Hb* is the percent of hemoglobin in the blood that is bound with O_2 . Figure 1 shows the relationship between percent saturation of Hb and P_{O_2} at normal body conditions.

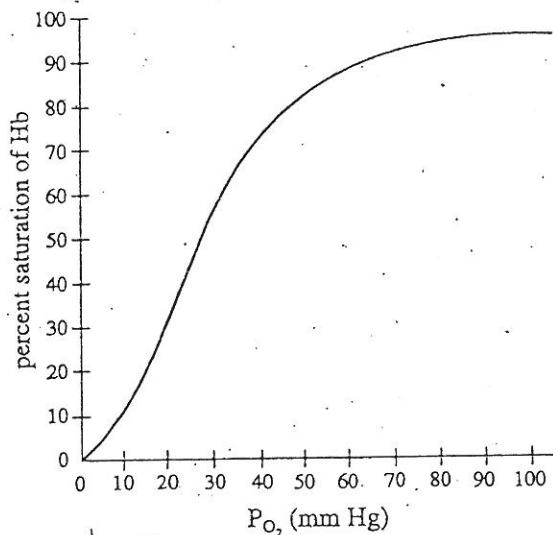


Figure 1

Figure 2 shows the relationship between percent saturation of Hb and P_{O_2} at different pHs. The pH of blood is affected by the partial pressure of CO_2 (P_{CO_2}).

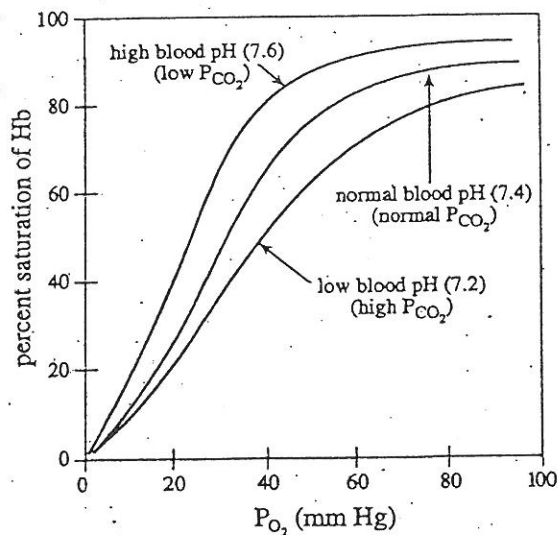


Figure 2

Figure 3 shows the relationship between percent saturation of Hb and P_{O_2} at different blood temperatures.

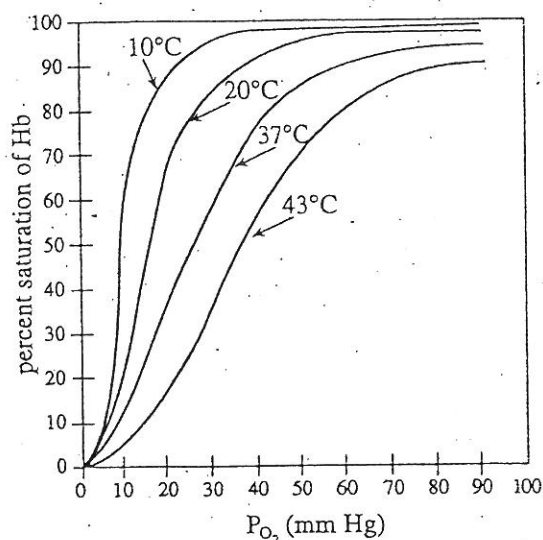


Figure 3

Figures adapted from Robert Carola, John P. Harley, and Charles R. Noback, *Human Anatomy and Physiology*. ©1990 by McGraw-Hill, Inc.

36. According to Figure 1, the greatest change in the percent saturation of Hb occurs within which of the following partial pressure ranges?

F. 0 mm Hg to 10 mm Hg
 G. 20 mm Hg to 30 mm Hg
 H. 40 mm Hg to 50 mm Hg
 J. 60 mm Hg to 70 mm Hg

37. Based on Figures 2 and 3, if blood has a P_{O_2} of 40 mm Hg, the amount of oxygen bound to Hb would be greatest at which of the following sets of conditions?

- A. Temperature = 10°C; pH = 7.6
- B. Temperature = 10°C; pH = 7.2
- C. Temperature = 43°C; pH = 7.6
- D. Temperature = 43°C; pH = 7.2

38. A researcher has claimed that the ability of Hb to bind with O_2 at a given P_{O_2} will decrease as temperature is increased. Do the data in Figure 3 support his claim?

- F. No; as the temperature increased, the percent saturation of Hb increased.
- G. No; as the temperature increased, the percent saturation of Hb decreased.
- H. Yes; as the temperature increased, the percent saturation of Hb increased.
- J. Yes; as the temperature increased, the percent saturation of Hb decreased.

39. Which of the following tables represents the assumptions about temperature and pH that would have to be made to properly interpret the data in Figures 1–3?

A.

	Factors held constant
Figure 1	pH only
Figure 2	temperature only
Figure 3	temperature and pH

B.

	Factors held constant
Figure 1	pH only
Figure 2	temperature and pH
Figure 3	temperature only

C.

	Factors held constant
Figure 1	temperature and pH
Figure 2	pH only
Figure 3	temperature only

D.

	Factors held constant
Figure 1	temperature and pH
Figure 2	temperature only
Figure 3	pH only

40. Based on the data in Figure 2, which of the following statements best explains why P_{CO_2} affects the pH of blood? When CO_2 dissolves in blood, it:

- F. forms an acid.
- G. absorbs acids.
- H. breaks down to form O_2 .
- J. reacts with O_2 to form Hb.

END OF TEST 4

STOP! DO NOT RETURN TO ANY OTHER TEST.

Form 60E (April 2005)
Parachuting

<u>English</u>		<u>Math</u>		<u>Reading</u>	<u>Science</u>
1) D	44) H	1) D	44) F	1) C	1) B
2) H	45) C	2) J	45) A	2) H	2) H
3) D	46) F	3) D	46) K	3) A	3) C
4) F	47) B	4) J	47) B	4) G	4) F
5) C	48) J	5) B	48) H	5) C	5) A
6) F	49) A	6) F	49) E	6) F	6) H
7) B	50) G	7) C	50) H	7) D	7) B
8) G	51) A	8) H	51) E	8) G	8) G
9) D	52) G	9) D	52) K	9) B	9) D
10) H	53) B	10) F	53) A	10) F	10) F
11) B	54) J	11) A	54) K	11) B	11) B
12) H	55) C	12) K	55) D	12) G	12) H
13) A	56) J	13) B	56) K	13) B	13) D
14) J	57) B	14) F	57) E	14) J	14) J
15) A	58) F	15) B	58) K	15) A	15) A
16) J	59) C	16) J	59) D	16) H	16) J
17) C	60) H	17) D	60) G	17) A	17) B
18) J	61) C	18) H		18) F	18) H
19) C	62) F	19) E		19) D	19) D
20) G	63) C	20) H		20) G	20) J
21) C	64) G	21) B		21) B	21) D
22) F	65) A	22) F		22) J	22) G
23) D	66) H	23) C		23) D	23) A
24) F	67) D	24) G		24) H	24) G
25) A	68) F	25) B		25) A	25) B
26) H	69) A	26) G		26) G	26) H
27) B	70) J	27) B		27) A	27) C
28) J	71) D	28) H		28) J	28) F
29) B	72) H	29) C		29) C	29) A
30) J	73) B	30) H		30) J	30) G
31) B	74) J	31) D		31) B	31) C
32) J	75) B	32) J		32) J	32) H
33) A		33) C		33) C	33) B
34) G		34) H		34) J	34) J
35) D		35) E		35) A	35) A
36) F		36) G		36) H	36) G
37) C		37) D		37) B	37) A
38) J		38) F		38) H	38) J
39) A		39) E		39) D	39) D
40) J		40) J		40) G	40) F
41) C		41) B			
42) J		42) F			
43) B		43) A			

ACT Resource Links

ACT Online Practice Tests: <https://www.crackab.com/act/>

✧ **ACT English Practice Tests:**

<https://www.crackab.com/act/english/>

✧ **ACT Math Practice Tests:**

<https://www.crackab.com/act/math/>

✧ **ACT Reading Practice Tests:**

<https://www.crackab.com/act/reading/>

✧ **ACT Science Practice Tests:**

<https://www.crackab.com/act/science/>

ACT Grammar: <https://www.crackab.com/act/grammar/>

ACT Real Past Papers Download:

<https://www.crackab.com/act-downloads/>

Digital SAT & New SAT Practice Tests:

<https://www.cracksat.net>

Real SAT Tests Download:

<http://www.cracksat.net/sat-downloads/>

AP Exams Practice Tests:

<https://www.crackap.com>

<https://www.apstudy.net>

Explanation of Procedures Used to Obtain Scale Scores from Raw Scores

On each of the four tests on which you marked any responses, the total number of correct responses yields a raw score. Use the table below to convert your raw scores to scale scores. For each test, locate and circle your raw score or the range of raw scores that includes it in the table below. Then, read across to either outside column of the table and circle the scale score that corresponds to that raw score. As you determine your scale scores, enter them in the blanks provided on the right. The highest possible scale score for each test is 36. The lowest possible scale score for any test on which you marked any responses is 1.

Next, compute the Composite score by averaging the four scale scores. To do this, add your four scale scores and divide the sum by 4. If the resulting number ends in a fraction, round it off to the nearest whole number. (Round down any fraction less than one-half; round up any fraction that is one-half or more.) Enter this number in the blank. This is your Composite score. The highest possible Composite score is 36. The lowest possible Composite score is 1.

ACT Test 60E

Your Scale Score

English

Mathematics

Reading

Science

Sum of scores

Composite score (sum ÷ 4)

NOTE: If you left a test completely blank and marked no items, do not list a scale score for that test. If any test was completely blank, do not calculate a Composite score.

Scale Score	Raw Scores				Scale Score
	Test 1 English	Test 2 Mathematics	Test 3 Reading	Test 4 Science	
36	75	60	40	40	36
35	74	--	39	39	35
34	73	59	37-38	38	34
33	72	58	36	--	33
32	71	57	35	37	32
31	70	55-56	34	36	31
30	69	53-54	33	--	30
29	67-68	51-52	31-32	35	29
28	65-66	49-50	30	33-34	28
27	63-64	46-48	28-29	32	27
26	61-62	43-45	27	30-31	26
25	59-60	41-42	26	29	25
24	57-58	38-40	25	27-28	24
23	54-56	36-37	23-24	25-26	23
22	52-53	34-35	22	24	22
21	49-51	32-33	21	22-23	21
20	46-48	30-31	20	20-21	20
19	43-45	28-29	19	18-19	19
18	41-42	25-27	18	16-17	18
17	38-40	22-24	17	14-15	17
16	35-37	18-21	16	13	16
15	32-34	15-17	15	12	15
14	29-31	12-14	13-14	11	14
13	27-28	09-11	11-12	10	13
12	25-26	07-08	09-10	09	12
11	23-24	06	07-08	08	11
10	20-22	05	06	07	10
9	18-19	04	05	05-06	9
8	15-17	03	--	04	8
7	12-14	--	04	--	7
6	10-11	02	03	03	6
5	08-09	--	--	02	5
4	06-07	01	02	--	4
3	04-05	--	--	01	3
2	02-03	--	01	--	2
1	00-01	00	00	00	1